

Gitea Sprint Handbook and Implementation Update

Beginner-friendly tasks, database contracts and handovers

CURRENT IMPLEMENTATION

React + Vite -> Express -> Prisma 7 -> Neon PostgreSQL | Gitea Actions

Member 6 branch member6-database

Verified commit 0227a28 | 36/36 local automated tests passing

Remote CI remains queued while the lecturer-managed Actions runner is unavailable.

COMS3011A Project 1 | 2026

Implementation update prepared 10 August 2026. Original sprint planning pages are preserved after page 9.

AI declaration: this revision was generated and reviewed with Codex[GPT-5]. The team must verify it before submission.

Implementation status - read this first

This section records what is actually present on the Gitea branch as at 10 August 2026. It overrides older planning-only wording where the two conflict.

Repository platform	Gitea at sdp.ms.wits.ac.za. Read older references to GitHub issues, PRs and Actions as Gitea issues, pull requests and Actions for this project.
Member 6 branch	member6-database on Gitea. Do not merge or delete it until review is complete.
Verified commit	0227a28 - feat: complete Sprint 2 Member 6 backend work, including Assisted-by: Codex[GPT-5].
Database stack	Prisma 7.9.1 with ordered SQL migrations for PostgreSQL; Neon is the deployed database target. Browser code must call Express, never Neon directly.
Local evidence	36/36 tests pass. Coverage: 96.66% statements, 94.33% branches, 90.47% functions and 96.62% lines.
Remote evidence	Gitea Actions run is queued at 0 seconds because the lecturer-managed runner is reported offline/unavailable. This is not a failed test result.

Status legend used in this update

Delivered: code and local tests exist. **Continue now:** teammates can code, test, push and review locally.

Dependency-owned: another member, coordinator or admin must provide something. **Blocked:** remote proof cannot be produced while the lecturer runner is offline.

Rules that remain unchanged

- Automated tests are mandatory. An offline runner does not remove the requirement to write and run them locally.
- Keep secrets out of Git: no .env, DATABASE_URL, Auth0 Management token, Codecov token or real personal data.
- Only the Integration Lead merges to develop or main; every member works on a branch and opens a small pull request.
- AI-generated code commits must include the required Assisted-by: <tool>[<model>] footer and README declaration where applicable.

Database delivered by Member 6

The database now supports the Basic organiser, tutor and student workflows. Teammates should build against these names instead of creating duplicate tables or bypassing the API.

Area	Models and purpose	Important states / rules
Identity and courses	Profile, Course, TutorMark, TimeSlot, TutorHourLimit	Roles are exactly ORGANISER, TUTOR, STUDENT. Marks are 0-100. Time ranges must end after they start.
Allocation	Allocation links one tutor to one course, date/time and allocated hours.	PROPOSED -> CONFIRMED -> CANCELLED is controlled. Hours must be greater than 0 and at most 24.
Tutor work	WorkLog, Timesheet, Excuse	Timesheet: DRAFT, SUBMITTED, APPROVED, RETURNED. Excuse: PENDING, APPROVED, REJECTED. Returned/rejected records require reasons.
Student open work	OverflowWork, VolunteerClaim	Work: OPEN, FILLED, CLOSED, CANCELLED. Claim: PENDING, ACCEPTED, REJECTED, WITHDRAWN. One student cannot claim the same item twice; one accepted claim per item.

Database-enforced protections

- Primary keys, foreign keys, uniqueness and compound ownership keys prevent orphan or cross-tutor records.
- Role triggers reject a student used as a tutor, a non-student volunteer claimant, or a non-organiser timesheet reviewer.
- State-transition triggers reject invalid jumps, such as approving a draft timesheet directly.
- Work logs may only be edited while the related timesheet is draft/returned; weekly work-log totals and hour ranges are enforced.
- Indexes support allocation candidates, dashboards, review queues, weekly hours and open-work boards.

Safe, repeatable demo data

Run `npm run db:seed`. The seed is rerunnable and contains only reserved *.example emails and demo|... Auth0 IDs. It provides one organiser, one student, three tutors and DEM0101 cases for pass, low mark, schedule clash, hour-limit, open work and closed work. Do not replace it with real student data.

Member 6 ownership boundary

Member 6 owns schema, migrations, seed, constraints, query helpers and CI configuration. Member 5 owns authenticated Express routes, application validation, transactions and safe error responses. Frontend members call Member 5's API; they do not import Prisma or SQL.

Stable query and adapter contracts

These implemented helpers are ready for Member 5 to wrap in authenticated Express routes. Route URLs remain Member 5's responsibility and must be agreed before frontend mocks are replaced.

Helper	Input	Safe output / meaning
findAllocationCandidates	courseId, startsAt, endsAt, minimumMark, requestedHours	For each tutor: tutorId, fullName, mark, weeklyLimit, usedHours, remainingHours, hasClash, isEligible, reasonCodes.
Candidate reasons	Calculated from the requested allocation	Stable codes: insufficient-mark, schedule-clash, hour-limit-exceeded. Member 2 maps these to friendly UI text.
getTutorHourSummary	tutorId and a Monday UTC weekStart	tutorId, weeklyLimit, usedHours, remainingHours, or null when no tutor record matches.
listOpenWork	Current time; defaults to now	Only future, unexpired OPEN items: id, courseCode, courseName, startsAt, endsAt, hours, status, closesAt. No tutor/student identity is exposed.

External-service adapter already implemented

src/integrations/external-api.ts provides backend-only timeout cancellation, strict JSON decoding into an approved safe shape, a source timestamp, and controlled fallbacks. The fallback reason is one of timeout, rate-limited, unavailable, invalid-response, http-error or network-error.

```
success: { source: "external", data: SAFE_DATA, retrievedAt: ISO_TIMESTAMP }
fallback: { source: "fallback", data: SAFE_DEFAULT, retrievedAt: null,
  reason: "timeout" | "rate-limited" | ... }
```

Not yet an HTTP API

The branch currently exposes only /health and /ready. Member 5 must add authentication, role/ownership checks, request validation, transactions and HTTP routes around these helpers. A domain-specific external service must not be invented until the coordinator/client approves its purpose, public fields, privacy/limit decision and freshness rule.

Files to use - do not copy SQL into frontend code

- Schema: prisma/schema.prisma; ordered migrations: prisma/migrations/; seed: prisma/seed.sql.
- Query helpers: src/services/allocation-queries.ts; database client: src/database.ts.
- Adapter: src/integrations/external-api.ts; handoff notes: docs/member-6-handoff.md.

What every member can do now

The runner does not prevent coding, local testing, pushing branches, reviewing code or opening pull requests. Use the Member 6 contracts below to reduce waiting and merge conflicts.

Owner	Continue now	Handoff / dependency
Member 1 - Auth/security	Keep Auth0 subject in Profile.auth0Id. Publish bearer-token, role and ownership middleware rules for the exact database roles. Add local route-guard tests.	Give Member 5 the verified token claims and 401/403 behavior. Database role triggers protect relationships but do not replace authentication.
Member 2 - Organiser	Build course/tutor/allocation board against mock JSON matching page 4. Display all three candidate checks and stable reason codes. Add UI tests for pass/fail/edit/remove.	Member 5 must expose candidate and allocation endpoints. Refresh hours and allocations after a successful transaction.
Member 3 - Tutor	Build work-log, weekly-hours, timesheet and excuse journeys using the exact statuses on page 3. Test loading, empty, invalid transition and returned/rejected reason states.	Member 5 must enforce the logged-in tutor's ownership and call database transactions. Member 3 never sends a tutor ID that can override the authenticated identity.
Member 4 - Student	Build the open-work board from the safe projection on page 4 and claim status UI. Test duplicate claim and closed/expired item responses. Prepare external success/fallback components.	Member 5 must expose open-work and claim routes. External live data waits for coordinator/client service approval, but the fallback UI can be built now.
Member 5 - Express backend	Wrap Member 6 queries in authenticated routes, add validation, transactions and safe errors. Add Supertest tests for every 200/400/401/403/404/409/500 path used.	Coordinate payloads with Members 1-4. Use Prisma/SQL only on the server. Do not weaken database constraints to make a route pass.
Member 6 - Data/CI	Maintain migrations, seed, queries, adapter and workflows; review database contract questions. Provide local test and coverage evidence.	Remote Gitea run and Codecov proof wait for the lecturer runner. Domain-service wiring waits for coordinator/client approval and a Member 5 route.

Parallel-work rule

Frontend members should keep approved mock handlers until the matching Member 5 route and response test exist. Replace one mock at a time. This allows the team to continue even while backend routes and remote CI are incomplete.

Shared review rule

A pull request is not complete only because it was pushed. It needs a linked issue, local test command and result, screenshots/API examples where relevant, no secrets, AI attribution where applicable, the assigned reviewer and an honest note that remote Actions evidence is unavailable while the runner is offline.

Member 5 backend integration checklist

This page turns Member 6's data layer into a safe Express API without committing the team to unreviewed route names. Member 5 should publish the final OpenAPI/JSON examples before Members 2-4 remove their mocks.

Step	Required backend work	Proof before handoff
1. Authenticate	Verify Auth0 access token; map token subject to Profile.auth0Id; reject missing/invalid token with 401.	Supertest cases for missing, expired/invalid and valid token. Never trust role or profile ID from the browser.
2. Authorise	Check exact role plus record ownership on every route. Organiser reviews use an organiser profile; tutor/student actions use the logged-in profile.	403 tests for wrong role and cross-user access; 404 where hiding record existence is safer.
3. Validate	Validate IDs, ISO dates, Monday week starts, mark 0-100, requested/allocated hours >0 and <=24, and non-empty reasons.	400 responses use stable field/code/message objects; boundary tests cover exact limits.
4. Query	Call findAllocationCandidates, getTutorHourSummary and listOpenWork through the server-side query client.	Response examples match page 4 and expose no database URL, SQL, stack trace or unrelated identity.
5. Mutate safely	Use Prisma transactions for create/edit/remove allocation, submit/review timesheet, excuse review and volunteer-claim acceptance.	409 for constraint/state conflicts; totals refreshed after commit; rollback test for a failed multi-write operation.
6. Hand off	Publish route table, request/response JSON, auth/role matrix, reason codes and failure examples.	Members 2-4 can replace one mock each; Member 6 checks schema/constraint assumptions.

Suggested route areas - names must be agreed by the team

```

organiser: courses, tutors, allocation candidates, allocation create/edit/remove
tutor:     weekly hours, work logs, timesheets, excuses
student:   open work, volunteer claim create/withdraw
shared:    GET /health and GET /ready already exist

```

Do not bypass the architecture

React must never import Prisma, execute SQL or receive DATABASE_URL. The only supported flow is React -> authenticated Express route -> Prisma/query helper -> Neon PostgreSQL -> safe JSON.

Frontend integration examples for Members 2-4

These examples match the implemented query layer. Member 5 may wrap them in a common envelope, but must not rename fields without first updating mocks, tests and this handoff.

Member 2 - allocation candidate card

```
{
  "tutorId": 12, "fullName": "Demo Tutor Clash", "mark": 72,
  "weeklyLimit": 8, "usedHours": 7, "remainingHours": 1,
  "hasClash": true, "isEligible": false,
  "reasonCodes": ["schedule-clash", "hour-limit-exceeded"]
}
```

Show each failed rule next to the tutor. Disable save when isEligible is false, but still let the organiser read every reason. Do not recalculate the final decision only in React.

Member 3 - weekly hours and workflow states

```
{ "tutorId": 10, "weeklyLimit": 10, "usedHours": 6,
  "remainingHours": 4 }
```

Use the server value after every successful work/allocation change. Timesheet states are DRAFT, SUBMITTED, APPROVED, RETURNED; excuse states are PENDING, APPROVED, REJECTED. Returned/rejected UI must show the review reason.

Member 4 - student-safe open-work item

```
{
  "id": 4, "courseCode": "DEM0101", "courseName": "Demo Course",
  "startsAt": "2030-09-04T14:00:00.000Z",
  "endsAt": "2030-09-04T16:00:00.000Z", "hours": 2,
  "status": "OPEN", "closesAt": null
}
```

Render only future, unexpired OPEN work. Do not expect tutor or student IDs in this list. After a claim, refresh the item and show PENDING, ACCEPTED, REJECTED or WITHDRAWN. A duplicate claim should produce a controlled conflict message.

Mock-to-live replacement test

Before removing a mock, compare the live response keys and state values with this page, then run the member's component tests plus Member 5's matching API test locally. Record any contract difference in the pull request instead of silently adapting only one side.

Blocked remote work - lecturer Actions runner offline

Red items below are the work the team cannot complete remotely until the lecturer-managed Gitea Actions runner is available. They are external blockers, not proof of a code defect.

BLOCKED: Gitea Actions jobs cannot start

The branch run is queued at 0 seconds. No member can produce a green or red remote job result, download Action logs, or prove that the workflow executed on Gitea while no runner accepts the job.

BLOCKED: remote CI and Codecov evidence

The backend quality job, PostgreSQL 17 migration/seed/verification job, API smoke job and Codecov upload cannot execute remotely. Therefore the team cannot truthfully tick 'CI and Codecov checks are green' yet.

BLOCKED: required passing-check merge gate

A required Gitea Actions check cannot pass while the runner is offline. Do not weaken tests, mark a queued run as passed, or claim branch protection evidence that does not exist. The repository administrator can require the checks after the runner is functioning.

What the lecturer's 7 August 2026 email means

The lecturer created organisation space, instructed the team to create its repository, required automated tests, and stated that an Action runner would be set up so CI/CD could run those tests. The team's repository and tests exist; runner provisioning/availability remains outside the members' control.

Continue now - these tasks are not blocked

Do now	Evidence to attach to each pull request
Write and run automated tests locally; lint, type-check and build; push branches; open/review pull requests; fix real local failures.	Terminal output for the exact command, test count, coverage summary, relevant screenshots/API examples, commit hash, reviewer and a note: 'Remote Actions queued - lecturer runner unavailable.'
Member 6 local baseline	36/36 tests pass; build and repository security checks pass; coverage is above the configured backend thresholds.

Do not repeatedly restart queued jobs. Once the lecturer confirms the runner is online, rerun once, inspect every job and keep the first successful logs as remote evidence.

Verification, pull request and remaining handoff

Use this checklist to finish the Member 6 review and unblock the rest of the team without misrepresenting the unavailable remote runner.

Local verification sequence

```
npm ci
npm run generate
npm run check
npm run db:migrate
npm run db:seed
npm run db:verify
```

Use a safe local or disposable PostgreSQL database for the last three commands. Never run `db:reset` against shared or production data. Member 6's in-memory migration tests are already part of `npm test`.

Pull request item	Current instruction
Source branch	member6-database
Target branch	develop - do not push Member 6 work directly to main.
Reviewer	Request Member 4, matching S2-M6-1/2/3 planning. Member 5 should also confirm the query/transaction handoff before using it.
Merge owner	Integration Lead only, after review and after any team rule for the offline runner is documented.
Attach	Commit 0227a28; 36/36 tests; coverage summary; build/security result; schema/seed/query summary; screenshot of queued Action; blocker wording from page 8.

Dependency-owned follow-up - not part of the runner blocker

- Coordinator/client approves the external service, purpose, public fields, privacy/limit decision and freshness/cache rule. Then Member 5 exposes it through an authenticated route and Member 4 renders the safe success/fallback shape.
- Repository administrator sets `CODECOV_TOKEN`, and later `DEPLOYMENT_URL` after an HTTPS backend exists. Frontend owner uploads LCOV using the reserved frontend flag.
- After the runner returns, require Backend quality and coverage and Clean PostgreSQL migration and API smoke on protected integration branches.

Original planning pages follow

Pages 10-103 preserve the earlier sprint issue library and review rubrics. Use this implementation update for the current branch name, Gitea terminology, database contracts, completed Member 6 work and runner status. Planning checkboxes that require green CI remain unticked until the runner actually executes the jobs.

AI declaration: this revision was generated and reviewed with Codex[GPT-5]. The project team remains responsible for checking the database contracts, lecturer instructions and all submitted evidence.

How to use this handbook

START HERE - 10 MINUTES

1) Decide which real person is Member 1-6. 2) Decide the Integration Lead (TBD). 3) Create develop and protect develop/main. 4) The Project Coordinator copies each issue into GitHub and assigns it. 5) Each Member works only on the branch printed in their issue, opens a PR to develop and waits for review. 6) Integration Lead merges and runs the release gate.

Who does what

Member 1	Authentication, role security, notifications/search and shared release checks	Yes - 3 issues x 4 points
Member 2	Organiser interface, allocation board, approvals, reports and scenarios	Yes - 3 issues x 4 points
Member 3	Tutor dashboard, availability, timesheets, excuses, imports and swaps	Yes - 3 issues x 4 points
Member 4	Student overflow plus shared responsive and accessible interface	Yes - 3 issues x 4 points
Member 5	Separate handwritten backend HTTP API, rules, security and API tests	Yes - 3 issues x 4 points
Member 6	Supa Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery CI/Co	Yes - 3 issues x 4 points
Project Coordinator (you)	Sprint meetings, GitHub board administration, stakeholder/user notes, decisions and all project/documentation-site writing.	No - deliberately excluded
Integration Lead (TBD)	Only person who integrates approved PRs and opens develop -> main release PR. Authors still resolve their own conflicts/defects.	Release gate; not a substitute for Member work

FAIRNESS NOTE

All six Members receive exactly 12 technical story points per sprint. The Coordinator and Integration Lead duties are excluded from those technical points. If those duties become large enough to affect fairness, the group must record a rebalancing decision before the sprint starts; do not hide extra work.

Plain-English terms

User story	A short sentence naming who needs something, what they need and why.
Acceptance test	A checkable Given/When/Then outcome. It says what must be true, not how to code it.
Task checkbox	A small implementation step the assignee ticks in the GitHub issue.
Branch	A safe line of work. One issue uses one named branch.
Pull request (PR)	A request to review and merge a branch into develop.
develop	Shared integration branch. Every Member PR targets this first.
main	Protected release branch. Only the Integration Lead merges a tested release into it.
Coverage	The code executed by automated tests. It is evidence, not proof that the app is correct.
Codecov	Receives coverage files made by the test runner and reports changes in GitHub.
Migration	A repeatable versioned database change; do not edit production structure manually.
Handwritten API	HTTP endpoints the team writes in Express. React calls these endpoints; Neon stores data but never replaces the application API.

GitHub setup and branch flow

RECOMMENDED BRANCH MODEL

Member issue branch -> pull request -> protected develop -> release pull request -> protected main.
Members never push directly to develop or main. The Integration Lead merges only after the issue acceptance tests, assigned review, CI and Codecov are green.

Repository settings checklist

- [] Create main and develop.
- [] Protect both branches: require a pull request, one approval, resolved conversations and required status checks.
- [] Block force pushes and deletion. Do not let the Integration Lead bypass failed checks.
- [] Create GitHub labels: sprint-1..4, member-1..6, points-4, area-*, type-test, bug and blocked.
- [] Create milestones for the four official dates.
- [] Create a GitHub Project board with Backlog, Ready, In progress, Review, Blocked and Done.

Windows PowerShell / Visual Studio Code terminal: work on an issue

```
# Open the repository folder in the VS Code terminal
git checkout develop
git pull origin develop
git checkout -b sl/ml/auth-entry      # replace with the exact issue branch

# Make small changes, then run the repository's tests
git status
git add .
git commit -m "feat(auth): implement sign-up and sign-in"
# If AI generated code, add a blank line and:
# Assisted-by: <tool>[<model>]

git push -u origin sl/ml/auth-entry
# On GitHub: open PR into develop, add "Closes #ISSUE_NUMBER",
# assign the printed reviewer, and wait for green checks.
```

If develop changed while you worked

```
git checkout develop
git pull origin develop
git checkout sl/ml/auth-entry
git merge develop
# Resolve only files you understand, rerun tests, then:
git add .
git commit -m "chore: resolve develop integration conflicts"
git push
```

DO NOT COPY THIS MISTAKE

Do not create one permanent Member branch for the whole project. Use the exact issue branch printed in this handbook. Small PRs are easier to review, test, revert and trace to a user story.

GitHub sprint method

The handbook uses a beginner-friendly Scrum-style sprint cycle. The official brief requires a motivated project methodology; this team process must be used consistently, not only described.

Before the sprint - Project Coordinator

- [] Choose the Sprint Goal and confirmed tier scope.
- [] Copy the 18 issue bodies for that sprint into GitHub; preserve the issue ID in every title.
- [] Add assignee, labels, milestone, four points and dependencies.
- [] Record client/user decisions and update public documentation. Members supply raw evidence, not documentation writing.

During the sprint - every Member

- [] Move issue Ready -> In progress only when starting it.
- [] Work on the exact issue branch and tick tasks as they are completed.
- [] Ask/block in GitHub when a dependency contract is missing; continue with the agreed mock when safe.
- [] Run local tests and coverage before opening the PR.
- [] Open one reviewable PR to develop, link the issue, respond to review and keep AI attribution correct.

Review and Done - reviewer plus Integration Lead

- [] Reviewer reproduces acceptance tests, checks code/test quality, permissions, secrets and scope.
- [] CI must pass and Codecov must show the correct coverage reports.
- [] Integration Lead merges approved PRs into develop, one at a time, and runs the full regression after integration.
- [] Issue moves to Done only when deployed/integrated behavior passes; code sitting on a branch is not Done.
- [] Only the Integration Lead opens and merges develop -> main after the sprint release gate.

Standard PR body - copy/paste

```
## Linked issue
Closes #<issue-number> (<stable ID such as S2-M3-2>)

## What changed
-

## Acceptance evidence
- [ ] Every acceptance test in the issue passes
- [ ] Local tests and coverage run
- [ ] CI and Codecov are green
- [ ] No secret/personal data added
- [ ] Screenshots/API examples/test output attached where useful
- [ ] AI attribution is correct

## Handoff
- Contract/schema/UI behavior another Member must know:
- Known limitation or follow-up issue:
```

Recommended architecture: React + Express + Neon

STACK DECISION FOR THIS EDITION

Supabase is removed from the Tutor Mx implementation plan. React + Vite builds the interface. Every Tutor Mx data request goes to the team's handwritten Node.js + Express API. The API uses Prisma to access Neon PostgreSQL. Auth0 is used only for authentication and identity.

Layer	What it does	Beginner rule
React + Vite	Pages, forms, role screens and calls to the Express API. Auth0 SDK handles login state.	Browser code never connects directly to Neon or Prisma.
Auth0	Sign-up, sign-in, sign-out, password reset and access tokens.	Authentication only. It does not become the Tutor Mx data API.
Node.js + Express	Handwritten HTTP routes, validation, role/ownership checks, business rules and external API adapter.	This is the application API the group must design and implement.
Prisma + Neon	Prisma runs in the backend; Neon hosts PostgreSQL tables, constraints, indexes and data.	DATABASE_URL stays server-side. Migrations are versioned in Git.
External API	One useful external integration called through a backend adapter.	Protect its secret, validate responses and keep a fallback.

Required request path

- 1) User signs in with Auth0 -> receives an access token
 - 2) React sends the request to OUR Express endpoint
 - 3) Express checks identity + role + ownership + validation
 - 4) Prisma queries Neon PostgreSQL
 - 5) Express returns safe JSON to React
- External API secret -> Express adapter only -> safe Tutor Mx response

Security checklist

- [] Never put DATABASE_URL or Auth0 Management API tokens in React/browser code.
- [] Express authorises every protected action; hidden buttons are not security.
- [] Prisma/database constraints still protect invalid relationships and hours.
- [] GitHub secrets hold deployment, Codecov and external API secrets.
- [] Seed/demo data is invented and non-personal.

Tests and Codecov on Windows / VS Code

IMPORTANT

Codecov does not run your tests. Your test runner creates a coverage report. GitHub Actions uploads that report to Codecov. The simplest beginner workflow is: run tests locally -> push the issue branch -> read the GitHub PR checks and Codecov comment.

Default npm-based frontend/backend commands

```
# From the repository folder in PowerShell
cd frontend
npm ci
npm run test:coverage
Test-Path .\coverage\lcov.info

cd ../backend
npm ci
npm run test:coverage
Test-Path .\coverage\lcov.info
```

CURRENT STACK - USE THESE COMMANDS

The backend for this handbook is Node.js + Express. The frontend is React + Vite. Use the npm commands above for both applications.

Do not switch the backend to .NET or Python halfway through the project unless the team records the decision and the tutor approves it. Keeping one agreed stack means one README, one CI setup and much less confusion.

Backend check in VS Code / PowerShell:

```
cd backend
npm ci
npm run test:coverage
Test-Path .\coverage\lcov.info
```

How a Member checks Codecov

- [] Run local tests and confirm a coverage file exists.
- [] Push the issue branch and open the PR to develop.
- [] Open the PR's Checks tab and wait for frontend tests, backend tests, codecov/patch and codecov/project (or configured equivalents).
- [] If upload is missing, open the workflow log and check the file path; do not paste the Codecov token into chat, code or terminal output.
- [] A high percentage does not replace acceptance tests. Test important success, boundary, permission and failure behavior.

Codecov GitHub Actions setup

Install the Codecov GitHub App for the repository, then add the repository upload token as the GitHub Actions secret CODECOV_TOKEN. Do not include the text CODECOV_TOKEN= in the secret value.

Example workflow - adapt paths to the chosen repository

```
name: tests-and-coverage
on:
  push:
  pull_request:

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 22

      - name: Frontend tests
        working-directory: frontend
        run: |
          npm ci
          npm run test:coverage

      - name: Backend tests
        working-directory: backend
        run: |
          npm ci
          npm run test:coverage

      - name: Upload frontend coverage
        uses: codecov/codecov-action@v5
        env:
          CODECOV_TOKEN: ${ secrets.CODECOV_TOKEN }
        with:
          files: ./frontend/coverage/lcov.info
          flags: frontend

      - name: Upload backend coverage
        uses: codecov/codecov-action@v5
        env:
          CODECOV_TOKEN: ${ secrets.CODECOV_TOKEN }
        with:
          files: ./backend/coverage/lcov.info
          flags: backend
```

Suggested team coverage rule - not an official course threshold

- [] Start with project target auto so coverage does not quietly fall.
- [] Aim for strong patch coverage on new code (for example 80%) only after the baseline is stable.
- [] Never add meaningless tests solely to raise a number; acceptance, permission, boundary and failure behavior matter.
- [] If the team changes a threshold, the Project Coordinator records the reason and date.

CURRENT OFFICIAL REFERENCES

Codecov's official GitHub guide uses codecov/codecov-action@v5 and a CODECOV_TOKEN secret. GitHub protected branches can require approving reviews and successful status checks before merge. URLs are listed in the final sources section.

AI policy checklist

COURSE RULE

AI use is encouraged, but attribution is mandatory. A document must contain a use or non-use declaration. The person submitting work is responsible for every AI-generated claim, command and line of code.

For every Member and PR

- [] Before prompting: remove secrets, tokens and personal data.
- [] Read, understand and test AI-generated code before committing it.
- [] For a commit containing AI-generated code, add: Assisted-by: <tool>[<model>] (list every tool/model used).
- [] Inline editing/autocomplete and AI code review do not need per-commit attribution, but their tool/model use must be declared separately in the README.
- [] Tell the Project Coordinator when AI use changes so the README/document declarations remain correct.

For the Project Coordinator's documents and README

- [] Every submitted document includes either the required tool/model/purpose declaration or: The preceding document was written without the assistance of AI.
- [] README separately declares use or non-use for AI code generation, AI inline editing and AI code review.
- [] For an assessment where AI is used, start a new session and preserve the unedited transcript or required substitute.
- [] Keep evidence honest; AI errors/hallucinations remain the team's responsibility.

Commit example

```
git commit

feat(allocation): validate weekly hours before save

Assisted-by: ChatGPT-Work[GPT-5]

# If no AI generated code for this commit, omit Assisted-by.
# Do not invent tool/model names; use the actual tool and model.
```


Sprint 1: plan, order and handovers

SPRINT GOAL

3-25 August 2026

Project setup, blockers and one integrated vertical foundation

What this sprint must achieve

- Create the GitHub, Auth0, Neon PostgreSQL and CI foundations and prove one thin path from sign-in, through the handwritten Express API, to PostgreSQL.
- Build simple Organiser, Tutor and Student screens with clear loading, empty, success and error states.
- Agree schemas and API examples early so frontend members can work safely with mocks.

Equal technical allocation

MEMBER	TECHNICAL OWNERSHIP	ISSUES	POINTS
Member 1	Authentication, role security, notifications/search and shared release checks	S1-M1-1 to S1-M1-3	12
Member 2	Organiser interface, allocation board, approvals, reports and scenarios	S1-M2-1 to S1-M2-3	12
Member 3	Tutor dashboard, availability, timesheets, excuses, imports and swaps	S1-M3-1 to S1-M3-3	12
Member 4	Student overflow plus shared responsive and accessible interface	S1-M4-1 to S1-M4-3	12
Member 5	Handwritten backend HTTP API, validation, business rules, security and API tests	S1-M5-1 to S1-M5-3	12
Member 6	Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery	-1 to S1-M6-3	12
TOTAL	Meetings and documentation are owned by the Project Coordinator.	18 issues	72

BEGINNER RULE

A contract is a written example of the data, API request, response and error shape. Once the example is agreed, frontend work may begin with a mock. A member waits only for the exact missing contract - not for the whole backend or whole sprint.

Sprint 1: who goes first and how work joins

WHO STARTS FIRST

Members 6 and 1 can start immediately in parallel. Member 6 publishes the Neon/Prisma schema, seed data and CI rules. Member 1 configures Auth0 and publishes the role/token contract. Member 5 starts the Express API as soon as those contracts are usable. Members 2-4 begin with the approved mock JSON and do not wait for live endpoints.

Detailed order, parallel work and handover

STEP / WHEN	OWNER	DO THIS	SAFE PARALLEL WORK	HAND OVER TO	READY WHEN
1 Day 1	Member 6	Publish schema, seed data, migration command and CI/Codecov contract.	Member 1 drafts role names. Members 2-4 sketch screens and states.	Member 5 + all UI members	A teammate can set up the database and read example records.
2 After schema draft	Members 5 + 1	M5 defines API route/payload/error examples. M1 defines bearer token and role rules.	M6 improves CI and seed data. M2-4 build against agreed mock JSON.	Members 2-4	Contracts are posted in GitHub and examples match the schema.
3 Same time	Members 2-4	Build Organiser, Tutor and Student shells, including loading/empty/error states and UI tests.	Each member owns a different role screen, so they can work at the same time.	Member 5	Mock-based screens and tests pass without a live backend.
4 When route is ready	Member 5	Implement the thin handwritten API route with validation and role checks; publish a working request/response example.	M1 tests auth failures; M6 checks database queries and CI.	Members 1-4	API tests pass and the example can be copied exactly.
5 After API example	Members 1-4	Replace only the matching mocks, connect real calls and keep friendly error messages.	Each UI member integrates their own route; M5 fixes shared API defects.	Member 6	Frontend-to-API tests pass and browser code never reads application data directly from Neon/PostgreSQL.
6 Integration	Member 6	Run clean setup, migrations, CI, Codecov and one full sign-in-to-database smoke test.	Issue reviewers review small PRs while checks run.	Integration Lead	Required checks are green; blockers are recorded with an owner.
7 Finish	All + Integration Lead	Authors fix review comments. Reviewers re-test. Integration Lead merges approved PRs to develop one at a time.	Next safe issue may start while an unrelated PR is reviewed.	Project Coordinator	Coordinator receives links, screenshots and test evidence for the sprint review.

How to hand over: post the contract or PR link in the dependent GitHub issue; include setup steps, sample data/request/response, tests run and known limits; tag the named receiver. The receiver replies that they can use it or opens a blocking issue naming the missing item. A handover is not complete just because code was pushed.

Sprint 1 - Member 1

YOUR SPRINT OWNERSHIP

Authentication, role security, notifications/search and shared release checks

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S1-M1-1 | Auth0 sign-up, sign-in and sign-out

GitHub issue ID	S1-M1-1	Story points	4
Branch	s1/m1/auth-entry	PR reviewer	Member 2
Labels	sprint-1, member-1, area-auth, points-4	PR base	develop
Depends on	Auth0 application settings and frontend shell	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a new or returning user, I want to create an account, sign in and sign out so that I can enter Tutor Mx securely.

Acceptance tests

- [] Given a valid new email and password, when the user signs up, then Auth0 creates the account and the app shows a clear next step.
- [] Given valid credentials, when the user signs in, then a session is created and the correct role entry page opens.
- [] Given invalid credentials or an unavailable service, when sign-in fails, then the user sees a safe message and no session is created.
- [] Given a signed-in user, when they sign out, then protected pages can no longer be opened.

Tasks

- [] Create sign-up, sign-in and sign-out UI with labelled fields.
- [] Use the official Auth0 React SDK / Universal Login; Tutor Mx code must not store passwords.
- [] Add loading, success and error states and stop duplicate submissions.
- [] Add component tests for valid and invalid paths.
- [] Run local coverage and attach the result to the PR.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Give Member 5 the bearer-token shape and give Members 2-4 the signed-in user/role interface.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S1-M1-2 | Password reset, account deletion shell and role routing

GitHub issue ID	S1-M1-2	Story points	4
Branch	s1/m1/auth-lifecycle	PR reviewer	Member 2
Labels	sprint-1, member-1, area-auth, points-4	PR base	develop
Depends on	S1-M1-1 and role names Organiser/Tutor/Student	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a Tutor Mx user, I want to reset my password, request account deletion and enter the correct role area so that account access is safe and understandable.

Acceptance tests

- [] Given a known email, when reset is requested, then the user receives the configured reset path without revealing whether other accounts exist.
- [] Given an authenticated user, when deletion is requested, then a confirmation is required before the backend action is called.
- [] Given each role, when sign-in succeeds, then only the matching role navigation is shown.
- [] Given a user without a permitted role, when they open a protected route, then access is blocked with a safe message.

Tasks

- [] Build reset-request and new-password screens.
- [] Build a two-step delete-account confirmation; backend deletion may remain a documented stub in Sprint 1.
- [] Create a role-route map for the three roles.
- [] Add tests for route guards and cancel/confirm states.
- [] Never place the Auth0 Management API token or Neon DATABASE_URL in the browser.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Give Member 5 the role matrix and give the Project Coordinator screenshots/test evidence for the security guide.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S1-M1-3 | Authentication contract and permission tests

GitHub issue ID	S1-M1-3	Story points	4
Branch	s1/m1/auth-contract-tests	PR reviewer	Member 2
Labels	sprint-1, member-1, type-test, points-4	PR base	develop
Depends on	S1-M1-1, S1-M1-2 and S1-M5-2	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want one tested authentication contract so that every frontend feature sends the same token and the backend makes the final permission decision.

Acceptance tests

- [] Given no bearer token, when a protected endpoint is called, then it returns 401 in the agreed JSON format.
- [] Given a valid token with the wrong role, when a restricted endpoint is called, then it returns 403.
- [] Given a valid token and permitted role, when a protected endpoint is called, then the request reaches the handler.
- [] Given an expired or invalid token, when used, then no private data is returned.

Tasks

- [] Agree the Authorization: Bearer <token> contract with Member 5.
- [] Create reusable frontend request code that attaches the token.
- [] Add mocked frontend tests plus backend integration cases with Member 5 fixtures.
- [] Run npm run test:coverage in the affected app.
- [] Record any contract mismatch as a GitHub bug issue.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Member 5 consumes the token helper/fixtures; Members 2-4 use the request wrapper rather than copying auth code.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 1 - Member 2

YOUR SPRINT OWNERSHIP

Organiser interface, allocation board, approvals, reports and scenarios

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S1-M2-1 | Organiser allocation-board shell

GitHub issue ID	S1-M2-1	Story points	4
Branch	s1/m2/organiser-board-shell	PR reviewer	Member 1
Labels	sprint-1, member-2, area-organiser, points-4	PR base	develop
Depends on	Shared navigation from S1-M4-2; mock candidate contract from Member 5	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As an organiser, I want one allocation board showing courses, tutors and rule results so that I do not have to move through many forms.

Acceptance tests

- [] Given courses and tutors, when the board opens, then the organiser can select a course and see candidate rows on the same page.
- [] Given no data, loading or an error, when the board opens, then the state is explained and the page remains usable.
- [] Given a narrow screen, when the board is used, then the main action and rule results remain reachable.
- [] Given keyboard-only use, when controls are visited, then focus order and labels are understandable.

Tasks

- [] Create the organiser route and single-board layout.
- [] Add course selector, candidate list and allocation action placeholder.
- [] Use mock data matching Member 5's proposed response contract.
- [] Add loading, empty and error states.
- [] Add a component test for course selection and keyboard focus.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Give Member 5 a list of required fields and give Member 4 any reusable table/card needs.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S1-M2-2 | Course and tutor management UI

GitHub issue ID	S1-M2-2	Story points	4
Branch	s1/m2/course-tutor-ui	PR reviewer	Member 1
Labels	sprint-1, member-2, area-organiser, points-4	PR base	develop
Depends on	S1-M2-1 and S1-M5-2 contract	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want to capture and correct courses, tutors, marks, busy times and weekly-hour limits so that allocation checks use complete data.

Acceptance tests

- [] Given valid details, when a course or tutor form is submitted, then the UI sends the agreed request and shows success.
- [] Given missing or invalid values, when submitted, then field-level messages explain the correction.
- [] Given an edit or removal, when the organiser confirms it, then the UI updates without a stale row.
- [] Given a destructive action, when Cancel is chosen, then no request is sent.

Tasks

- [] Build course list/add/edit/remove components.
- [] Build tutor details form including marks, busy times and weekly-hour limit fields.
- [] Use mock handlers until Member 5 endpoints are ready.
- [] Add confirmation for removal and clear validation messages.
- [] Add tests for valid, invalid and cancelled actions.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Member 6 uses the final field list in migrations; Member 5 matches the request/response shapes.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S1-M2-3 | Immediate allocation-rule messages and UI tests

GitHub issue ID	S1-M2-3	Story points	4
Branch	s1/m2/allocation-rule-ui	PR reviewer	Member 1
Labels	sprint-1, member-2, type-test, points-4	PR base	develop
Depends on	S1-M2-1 and rule outputs from S1-M5-3	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want to see mark eligibility, timetable clashes and remaining weekly hours before saving so that I can correct an invalid choice immediately.

Acceptance tests

- [] Given a candidate below the mark threshold, when shown, then the mark rule visibly fails with a reason.
- [] Given overlapping busy time, when shown, then the clash rule visibly fails before save.
- [] Given insufficient remaining hours, when shown, then remaining and required hours are visible before save.
- [] Given all three checks pass, when the organiser selects the tutor, then the save action becomes available.

Tasks

- [] Create one reusable rule-result component for pass, fail and unknown.
- [] Show all three checks together; do not hide failed checks.
- [] Add tests for pass/fail/unknown and threshold boundary values.
- [] Check colour is not the only way status is communicated.
- [] Run local coverage and attach the summary to the PR.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Member 5 supplies stable reason codes; Member 2 maps those codes to plain language without duplicating business logic.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 1 - Member 3

YOUR SPRINT OWNERSHIP

Tutor dashboard, availability, timesheets, excuses, imports and swaps

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S1-M3-1 | Tutor dashboard shell

GitHub issue ID	S1-M3-1	Story points	4
Branch	s1/m3/tutor-dashboard	PR reviewer	Member 4
Labels	sprint-1, member-3, area-tutor, points-4	PR base	develop
Depends on	Role routing from S1-M1-2 and shared states from S1-M4-2	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a tutor, I want one dashboard showing what I owe and when so that I can understand my work immediately.

Acceptance tests

- [] Given assignments, when the dashboard opens, then course/session, date/time and expected hours are visible.
- [] Given no assignments, when the dashboard opens, then a clear no-work message is shown.
- [] Given slow or failed loading, when data is requested, then the page shows a usable status and retry action.
- [] Given another tutor's identifier, when a user tries to change the URL, then the UI does not reveal that tutor's work.

Tasks

- [] Build the tutor route and dashboard cards/table.
- [] Use mock data matching the planned own-dashboard endpoint.
- [] Add loading, empty, error and retry states.
- [] Display remaining weekly hours in a plain format.
- [] Add tests for filled, empty and error states.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Give Member 5 the dashboard fields and Member 6 the query/seed-data needs.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S1-M3-2 | Availability, work-log and excuse forms

GitHub issue ID	S1-M3-2	Story points	4
Branch	s1/m3/tutor-forms	PR reviewer	Member 4
Labels	sprint-1, member-3, area-tutor, points-4	PR base	develop
Depends on	S1-M3-1 and initial API contracts	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a tutor, I want simple forms for busy times, worked hours and excuses so that I can keep my availability and obligations accurate.

Acceptance tests

- [] Given valid busy time, when submitted, then start/end and recurrence placeholders are captured in the agreed format.
- [] Given valid worked hours, when submitted, then the date, session and hours are shown in a draft list.
- [] Given an assigned session and reason, when an excuse is submitted, then confirmation is shown.
- [] Given missing, negative or impossible values, when submitted, then no request is sent and the correction is explained.

Tasks

- [] Build availability/busy-time form.
- [] Build draft work-log form and list.
- [] Build session excuse form.
- [] Label every field and associate error text with the field.
- [] Add valid/invalid component tests.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Member 5 receives the payload shapes; Member 6 receives required tables and constraints.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S1-M3-3 | Tutor UI contract and accessibility tests

GitHub issue ID	S1-M3-3	Story points	4
Branch	s1/m3/tutor-ui-tests	PR reviewer	Member 4
Labels	sprint-1, member-3, type-test, points-4	PR base	develop
Depends on	S1-M3-1, S1-M3-2 and S1-M4-2	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want automated tutor-interface tests so that later API connection does not break the dashboard and forms.

Acceptance tests

- [] Given keyboard-only navigation, when the tutor journey is followed, then every control can be reached and activated.
- [] Given a common phone width, when the dashboard/forms render, then no required action is clipped.
- [] Given invalid hours or dates, when submitted, then the expected message appears.
- [] Given the mock contract changes unexpectedly, when tests run, then the mismatch fails the suite.

Tasks

- [] Write tests for dashboard, forms and shared states.
- [] Add one responsive rendering check at the agreed narrow width.
- [] Add one basic accessibility scan or explicit label/focus assertions.
- [] Generate frontend coverage locally.
- [] Fix uncovered critical branches in the tutor journey.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Give the Project Coordinator the test command/output and give Member 5 any contract failures as bugs.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 1 - Member 4

YOUR SPRINT OWNERSHIP

Student overflow plus shared responsive and accessible interface

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S1-M4-1 | Student overflow page shell

GitHub issue ID	S1-M4-1	Story points	4
Branch	s1/m4/student-overflow	PR reviewer	Member 3
Labels	sprint-1, member-4, area-student, points-4	PR base	develop
Depends on	Role routing from S1-M1-2	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a student, I want a plain list of unclaimed work so that I can quickly find something suitable to volunteer for.

Acceptance tests

- [] Given open work, when the page opens, then course, date/time, expected hours and availability status are visible.
- [] Given no open work, when the page opens, then a clear empty message is shown.
- [] Given slow or failed loading, when data is requested, then a status and retry action are shown.
- [] Given a closed item in mock data, when the list renders, then it cannot be volunteered for.

Tasks

- [] Build the student route and overflow list/cards.
- [] Use mock work items matching Member 5's planned contract.
- [] Add loading, empty, error and retry states.
- [] Keep the main volunteer action obvious on phone and desktop layouts.
- [] Add tests for open, closed, empty and error cases.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Give Member 5 and Member 6 the exact work-item fields needed by the UI.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S1-M4-2 | Shared role navigation and status components

GitHub issue ID	S1-M4-2	Story points	4
Branch	s1/m4/shared-navigation	PR reviewer	Member 3
Labels	sprint-1, member-4, area-shared-ui, points-4	PR base	develop
Depends on	Role map from S1-M1-2	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As any Tutor Mx user, I want consistent navigation and status messages so that I always know where I am and what happened.

Acceptance tests

- [] Given Organiser, Tutor or Student role, when the app opens, then only permitted navigation items are shown.
- [] Given loading, empty, error or success state, when used by any page, then wording and visual treatment are consistent.
- [] Given keyboard navigation, when focus moves, then the focused element is visible.
- [] Given a common phone width, when navigation opens, then no role's primary task is hidden.

Tasks

- [] Create role-aware navigation shell.
- [] Create reusable loading, empty, error, success and confirmation components.
- [] Add visible focus and skip-to-content support.
- [] Add responsive layout rules for shared shell.
- [] Add unit/component tests for all states and roles.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Members 2 and 3 consume these components; Member 1 supplies the authoritative role list.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S1-M4-3 | Volunteer confirmation and shared UI tests

GitHub issue ID	S1-M4-3	Story points	4
Branch	s1/m4/volunteer-confirmation	PR reviewer	Member 3
Labels	sprint-1, member-4, type-test, points-4	PR base	develop
Depends on	S1-M4-1 and S1-M4-2	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a student, I want to confirm a volunteer request and avoid duplicate clicks so that I do not claim the wrong work twice.

Acceptance tests

- [] Given an open item, when Volunteer is selected, then confirmation repeats the item details.
- [] Given Cancel, when confirmation closes, then no request is sent.
- [] Given Confirm, when a request is pending, then the button is disabled until success or failure.
- [] Given a repeated click or already-claimed item, when attempted, then the UI prevents or explains the duplicate.

Tasks

- [] Build the confirm/cancel interaction.
- [] Add pending, success and failure states using shared components.
- [] Add duplicate-submit protection.
- [] Write tests for confirm, cancel, pending and duplicate cases.
- [] Run local coverage and fix uncovered critical paths.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Member 5 receives expected duplicate/error responses; the Project Coordinator receives test evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 1 - Member 5

YOUR SPRINT OWNERSHIP

Separate handwritten backend HTTP API, rules, security and API tests

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S1-M5-1 | Separate handwritten backend and health endpoint

GitHub issue ID	S1-M5-1	Story points	4
Branch	s1/m5/backend-foundation	PR reviewer	Member 6
Labels	sprint-1, member-5, area-api, points-4	PR base	develop
Depends on	Chosen backend runtime and repository structure	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a developer, I want a separately runnable backend with a handwritten HTTP API so that the frontend never depends on direct database access or generated database endpoints.

Acceptance tests

- [] Given a clean checkout and environment variables, when the backend start command runs, then it starts without the frontend.
- [] Given GET /health, when the backend and database configuration are healthy, then a documented 200 JSON response is returned.
- [] Given an unhandled route or error, when requested, then a consistent safe error response is returned.
- [] Given the frontend codebase, when searched, then application data is not read/written through direct Neon/PostgreSQL client calls.

Tasks

- [] Create the backend application in a separate repository or clearly separate application.
- [] Add environment validation, request logging without secrets, and central error handling.
- [] Implement GET /health manually.
- [] Add lint, start and test scripts.
- [] Add health/error tests and generate coverage.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Give Member 6 the environment and database-connection requirements; give all frontend Members the base URL.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S1-M5-2 | Auth middleware plus profile, course and tutor endpoints

GitHub issue ID	S1-M5-2	Story points	4
Branch	s1/m5/core-api-contracts	PR reviewer	Member 6
Labels	sprint-1, member-5, area-api, points-4	PR base	develop
Depends on	S1-M5-1, auth contract from Member 1 and schema from Member 6	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the application, I want protected handwritten endpoints for profiles, courses and tutors so that role-safe data reaches the frontend through one API contract.

Acceptance tests

- [] Given a valid bearer token, when the backend verifies the Auth0 access token, then the user identity and role are available to handlers.
- [] Given no/invalid token, when a protected endpoint is called, then 401 is returned without private data.
- [] Given the wrong role, when organiser-only write endpoints are called, then 403 is returned.
- [] Given valid course/tutor data, when create/read/update requests are made, then the server validates and uses Neon PostgreSQL through Prisma/pg in backend code.

Tasks

- [] Implement token verification and role middleware.
- [] Implement profile/current-user read endpoint.
- [] Implement initial course and tutor read/write endpoints.
- [] Define one success/error JSON format and status-code table.
- [] Add endpoint tests for success, validation, 401 and 403.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Members 1-4 use the published route/payload examples; Member 6 aligns migrations and seed data.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S1-M5-3 | Three allocation-rule functions and boundary tests

GitHub issue ID	S1-M5-3	Story points	4
Branch	s1/m5/allocation-rules	PR reviewer	Member 6
Labels	sprint-1, member-5, area-allocation, points-4	PR base	develop
Depends on	Threshold decisions from client/coordinator and Member 6 data model	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want allocation rules checked consistently on the server so that marks, timetable clashes and weekly-hour limits cannot be bypassed.

Acceptance tests

- [] Given a mark at the agreed threshold, when eligibility is checked, then the documented inclusive/exclusive rule is applied consistently.
- [] Given touching and overlapping times, when clash is checked, then only the agreed cases fail with a reason code.
- [] Given used plus proposed hours at/exceeding the weekly limit, when checked, then remaining hours and pass/fail are correct.
- [] Given missing required data, when any rule runs, then it returns a controlled failure/unknown result rather than approving silently.

Tasks

- [] Create three pure domain functions independent of UI code.
- [] Return stable reason codes plus plain values needed by the UI.
- [] Write boundary tests for threshold, touching/overlap and exact/exceeded limits.
- [] Expose a draft validation endpoint or service interface.
- [] Run coverage and ensure every rule branch is exercised.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 2 maps reason codes to UI text; Member 6 provides candidate/availability/hour data.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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Sprint 1 - Member 6

YOUR SPRINT OWNERSHIP

Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery
3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S1-M6-1 | Neon PostgreSQL schema, Prisma migrations and safe seed data

GitHub issue ID	S1-M6-1	Story points	4
Branch	s1/m6/neon-schema		
Labels	sprint-1, member-6, area-database, points-4	PR base	develop
Depends on	Field lists from Members 1-5	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As the team, we want a repeatable Neon PostgreSQL schema so that every environment supports users, roles, courses, marks, busy times, weekly limits and allocations.

Acceptance tests

- [] Given an empty development database, when migrations run in order, then the required tables, keys and constraints are created.
- [] Given the safe seed command, when run, then linked Organiser/Tutor/Student demo data exists for all three allocation rules.
- [] Given invalid foreign keys or negative hours, when inserted, then database constraints reject them.
- [] Given repository search, when checked, then no DATABASE_URL, Auth0 Management token or production personal data is committed.

Tasks

- [] Create migrations for profiles/roles, courses, tutor marks, availability, hour limits and allocations.
- [] Add primary/foreign keys, uniqueness and basic value constraints.
- [] Create small non-personal seed data for three roles and rule boundaries.
- [] Add a reset/setup procedure that does not rely on manual production edits.
- [] Add migration/constraint tests or repeatable verification queries.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Member 5 consumes the backend database connection; Members 2-4 receive stable demo identifiers and fields.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S1-M6-2 | GitHub Actions, Codecov and protected integration flow

GitHub issue ID	S1-M6-2	Story points	4
Branch	s1/m6/ci-codecov	PR reviewer	Member 5
Labels	sprint-1, member-6, area-ci, points-4	PR base	develop
Depends on	Test scripts from Members 1-5 and repository admin access	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want every pull request tested and measured automatically so that the Integration Lead only merges work that keeps the shared build healthy.

Acceptance tests

- [] Given a pull request to develop, when it opens or changes, then install, lint and automated tests run.
- [] Given test coverage files, when CI completes, then Codecov receives frontend/backend reports and comments/checks on the PR.
- [] Given a failed test or required coverage status, when a merge is attempted, then branch protection blocks it.
- [] Given a secret, when CI runs, then it is read from GitHub Actions secrets and never printed or committed.

Tasks

- [] Create GitHub Actions workflow(s) for install, lint, tests and coverage.
- [] Install the Codecov GitHub App and add CODECOV_TOKEN as a repository secret.
- [] Upload coverage with codecov/codecov-action@v5 using frontend/backend flags.
- [] Configure develop and main to require PR review and passing checks.
- [] Document required check names for the Integration Lead via an issue comment.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

All Members use the same test script names; the Integration Lead uses these checks as the merge gate.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S1-M6-3 | Relevant external API spike and fallback contract

GitHub issue ID	S1-M6-3	Story points	4
Branch	s1/m6/external-api-spike	PR reviewer	Member 5
Labels	sprint-1, member-6, area-integration, points-4	PR base	develop
Depends on	Client/coordinator chooses a relevant service and confirms purpose	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a Tutor Mx user, I want one relevant external service to add useful information without breaking the core system when that service is unavailable.

Acceptance tests

- [] Given the chosen service and valid credentials, when a server-side spike runs, then one useful response is returned and recorded without exposing the secret.
- [] Given a timeout, unavailable service or invalid response, when the adapter runs, then a controlled fallback result is returned.
- [] Given the service's limits/privacy terms, when evaluated, then the coordinator can state why it is relevant and safe enough.
- [] Given no final service approval, when Sprint 1 ends, then a replacement decision and deadline are recorded.

Tasks

- [] Propose one domain-relevant API and a specific Tutor Mx use case.
- [] Call it from backend-side spike code, never directly with a secret in the browser.
- [] Add timeout, response validation and fallback interface.
- [] Add mocked success, timeout and invalid-response tests.
- [] Provide raw decision/test evidence to the Project Coordinator.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Member 5 will integrate the adapter into the handwritten API; Member 4 will later display the safe response/fallback.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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Sprint 1: integration review and official rubric

OFFICIAL DESCRIPTION NOTE

Weights and official level meanings come from project_briefs.pdf. The evidence wording below is condensed into Tutor Mx language for planning; use the official PDF when marks are disputed.

Integration Lead release gate

- [] Every issue is linked to a reviewed PR targeting develop; no Member pushes directly to main.
- [] The assigned reviewer checks the issue's acceptance tests, changed code, local/CI tests, coverage and AI attribution.
- [] Integration Lead merges approved PRs into develop one at a time, resolves conflicts with the author, then runs the full suite.
- [] Only the Integration Lead opens develop -> main. Main remains protected and merges only after required checks pass.
- [] The Project Coordinator runs meetings, maintains the GitHub board/decisions and writes project documentation from Member evidence.

Sprint review demonstration checklist

- [] Open GitHub Project and show all 18 issues, owners, points and linked PRs.
- [] Show protected develop/main, CI and Codecov checks.
- [] Run one role sign-in and one frontend -> handwritten Express API -> Neon PostgreSQL data trace.
- [] Show organiser board rule messages plus Tutor/Student shells.
- [] Show public documentation foundation and honest blockers/next sprint.

Official rubric weights

Version Control	10%	All members contribute; organised repository, code-quality tools and useful commits.
Documentation Site	10%	Public, accessible site with non-trivial project content.
Getting Started / Dev Guides	5%	Every repository has usable README/setup instructions.
Work Tracker	5%	GitHub tracker exists and is actively used.
Git Methodology	5%	Documented branch/PR method is actually followed.
Project Methodology	10%	Referenced, motivated sprint method with evidence of use.
Tech Stack	5%	Complete stack listed and motivated.
Stakeholder Interaction	10%	Requirements discussed and decisions evidenced.
Initial Design & Dev Plan	20%	Comprehensive design, UI, roadmap and development plan.
Implementation	20%	Supporting requirements and planned foundation features implemented.
TOTAL	100%	Official weights total 100%.

Sprint 2: plan, order and handovers

SPRINT GOAL

26 August-15 September 2026

Complete Basic journeys end to end and move toward stability

What this sprint must achieve

- Complete every Basic feature in the shared environment, not only in isolated branches.
- Deploy the handwritten API, finish the selected external integration and run useful UI/API tests through Codecov.
- Fix the representative-user and client findings selected by the Project Coordinator.

Equal technical allocation

MEMBER	TECHNICAL OWNERSHIP	ISSUES	POINTS
Member 1	Authentication, role security, notifications/search and shared release checks	S2-M1-1 to S2-M1-3	12
Member 2	Organiser interface, allocation board, approvals, reports and scenarios	S2-M2-1 to S2-M2-3	12
Member 3	Tutor dashboard, availability, timesheets, excuses, imports and swaps	S2-M3-1 to S2-M3-3	12
Member 4	Student overflow plus shared responsive and accessible interface	S2-M4-1 to S2-M4-3	12
Member 5	Handwritten backend HTTP API, validation, business rules, security and API tests	S2-M5-1 to S2-M5-3	12
Member 6	Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery	-1 to S2-M6-3	12
TOTAL		ues	72

BEGINNER RULE

A contract is a written example of the data, API request, response and error shape. Once the example is agreed, frontend work may begin with a mock. A member waits only for the exact missing contract - not for the whole backend or whole sprint.

Sprint 2: who goes first and how work joins

WHO STARTS FIRST

Members 1 and 6 start together: Member 1 freezes the Basic role/permission matrix while Member 6 publishes the Basic schema and migration changes. Member 5 starts endpoints as soon as each individual contract is ready. Members 2-4 continue role journeys with approved mocks in parallel.

Detailed order, parallel work and handover

STEP / WHEN	OWNER	DO THIS	SAFE PARALLEL WORK	HAND OVER TO	READY WHEN
1 Day 1	Members 1 + 6	M1 publishes Basic permissions and auth states. M6 publishes Basic schema, migrations and seed data.	They work independently but compare role names and record ownership before sign-off.	Member 5	Permission and data contracts use the same terms and examples.
2 Contract by contract	Member 5	Build Basic endpoints in small groups; document validation, errors and sample payloads as each group lands.	M6 deploys and monitors the API. M1 tests unauthorised access.	Members 2-4	Each route has passing API tests and a copyable request/response example.
3 Same time	Members 2-4	Finish Organiser, Tutor and Student Basic journeys with mocks; add loading, empty, error and retry states.	Role areas are separate, so all three members can build and test at once.	Member 5	Mock journeys pass and missing contracts are logged, not guessed.
4 As routes land	Members 2-4	Swap approved mocks for live endpoints one route at a time and run end-to-end tests.	M5 fixes API defects; M1 checks role boundaries; M6 watches deployment logs.	Issue reviewer	The full user journey passes in the shared environment.
5 After core journeys	Member 6 + relevant owner	Connect the selected external service through the backend and verify timeout/fallback behavior.	Other owners fix Coordinator-prioritised feedback and accessibility issues.	Member 5 + Member 4	Secrets stay server-side and the UI shows a safe fallback when the service fails.
6 Stability pass	All members	Run Basic regression, Codecov and representative-user checks; open one issue per reproducible defect.	Defects in different areas can be fixed in parallel when they do not touch the same contract.	Integration Lead	Severity-one/two defects are fixed or explicitly blocked with evidence.
7 Finish	Integration Lead	Merge reviewed PRs to develop one at a time, rerun regression, then open the protected release PR.	Coordinator prepares review evidence while final checks run.	Project Coordinator	Release is demonstrable end to end and evidence links are complete.

How to hand over: post the contract or PR link in the dependent GitHub issue; include setup steps, sample data/request/response, tests run and known limits; tag the named receiver. The receiver replies that they can use it or opens a blocking issue naming the missing item. A handover is not complete just because code was pushed.

Sprint 2 - Member 1

YOUR SPRINT OWNERSHIP

Authentication, role security, notifications/search and shared release checks

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S2-M1-1 | Complete Auth0 account lifecycle

GitHub issue ID	S2-M1-1	Story points	4
Branch	s2/m1/account-lifecycle	PR reviewer	Member 3
Labels	sprint-2, member-1, area-auth, points-4	PR base	develop
Depends on	Sprint 1 auth foundation and deployed backend	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a Tutor Mx user, I want sign-up, sign-in, sign-out, password reset and account deletion to work end to end so that I control my account safely.

Acceptance tests

- [] Given each required account action, when completed in the shared environment, then the expected result is visible and persists.
- [] Given deletion, when confirmed, then the backend performs the privileged action and the old session cannot access protected data.
- [] Given a failed/expired reset link, when opened, then a safe recovery path is shown.
- [] Given network failure, when any action runs, then the user is not left in a false signed-in/deleted state.

Tasks

- [] Connect every account screen to real Auth0 + handwritten backend behavior.
- [] Implement the server-only delete endpoint with Member 5; keep the Auth0 Management API token server-side.
- [] Add confirmations and post-action session cleanup.
- [] Add happy-path and failure integration tests.
- [] Generate coverage and fix uncovered critical account paths.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Members 2-4 receive a stable signed-in session/role API; Coordinator receives acceptance evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S2-M1-2 | Role protection across frontend and backend

GitHub issue ID	S2-M1-2	Story points	4
Branch	s2/m1/role-protection	PR reviewer	Member 3
Labels	sprint-2, member-1, area-security, points-4	PR base	develop
Depends on	S2-M1-1 and Member 5 Basic endpoints	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the school, we want role and ownership rules enforced on every request so that users cannot view or change data outside their responsibilities.

Acceptance tests

- [] Given an Organiser, when organiser-only endpoints/screens are used, then permitted actions work.
- [] Given a Tutor or Student, when organiser-only actions are attempted directly, then the backend returns 403.
- [] Given one tutor/student, when another user's record identifier is supplied, then private records are not returned or changed.
- [] Given hidden navigation, when the frontend is bypassed, then server authorisation still blocks the request.

Tasks

- [] Create an endpoint-to-role/ownership test matrix.
- [] Pair with Member 5 to apply middleware and ownership checks to all Basic endpoints.
- [] Update frontend route guards without treating them as the security boundary.
- [] Add 401, 403 and cross-user integration tests.
- [] Log safe permission failures without tokens/personal data.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Member 5 receives failing endpoint cases; Members 2-4 use the shared permission/error behavior.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S2-M1-3 | Cross-role security smoke suite and assigned fixes

GitHub issue ID	S2-M1-3	Story points	4
Branch	s2/m1/security-smoke	PR reviewer	Member 3
Labels	sprint-2, member-1, type-test, points-4	PR base	develop
Depends on	All Sprint 2 Basic journeys integrated on develop	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the Integration Lead, I want a repeatable cross-role smoke suite so that a merge cannot accidentally break account access or expose another role's data.

Acceptance tests

- [] Given clean demo accounts for all roles, when the smoke suite runs, then each user reaches only their expected landing page.
- [] Given expired/missing/wrong-role tokens, when critical endpoints are called, then 401/403 behavior is consistent.
- [] Given account deletion/reset, when completed, then subsequent access behaves correctly.
- [] Given a discovered severity-one/two security defect, when Sprint 2 closes, then it is fixed and re-tested or the release is blocked.

Tasks

- [] Create a scripted or automated cross-role smoke sequence.
- [] Run it against the shared environment after integration.
- [] Open GitHub bugs for failures with steps and severity.
- [] Fix the highest-priority account/security bugs assigned to Member 1.
- [] Attach passing test/coverage output to the PR.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Integration Lead runs this before the release PR; Coordinator links results into testing evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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Sprint 2 - Member 2

YOUR SPRINT OWNERSHIP

Organiser interface, allocation board, approvals, reports and scenarios

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S2-M2-1 | Real course and tutor CRUD on the organiser board

GitHub issue ID	S2-M2-1	Story points	4
Branch	s2/m2/organiser-crud	PR reviewer	Member 5
Labels	sprint-2, member-2, area-organiser, points-4	PR base	develop
Depends on	Member 5 organiser endpoints and Member 6 Basic schema	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As an organiser, I want to add, view, edit and remove courses/tutors on the same working area so that allocation data stays current.

Acceptance tests

- [] Given valid data, when create/edit is submitted, then the handwritten API stores it and the board refreshes.
- [] Given invalid data, when submitted, then server validation appears next to the correct fields.
- [] Given remove, when confirmed, then dependent-data rules are respected and the UI shows the result.
- [] Given API failure, when an action fails, then the previous correct screen state is preserved.

Tasks

- [] Replace Sprint 1 mocks with Member 5 endpoints.
- [] Complete create/read/update/delete UI for courses and tutors.
- [] Map server validation to field messages.
- [] Handle confirmations and dependent-data conflict responses.
- [] Add integration tests for success, validation and failure.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Member 5 fixes contract mismatches; Member 6 validates data constraints.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S2-M2-2 | Complete allocation create, edit and remove with live checks

GitHub issue ID	S2-M2-2	Story points	4
Branch	s2/m2/allocation-board-basic	PR reviewer	Member 5
Labels	sprint-2, member-2, area-allocation, points-4	PR base	develop
Depends on	S2-M2-1, Member 5 allocation endpoints and Member 6 queries	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want every tutor choice checked immediately and valid allocations saved from one board so that I can allocate confidently.

Acceptance tests

- [] Given a candidate, when selected, then mark, clash and remaining-hours results are shown before save.
- [] Given any failed rule, when save is attempted, then the backend rejects it and the reason remains visible.
- [] Given all rules pass, when saved, then the allocation appears without manual database editing.
- [] Given an existing allocation, when edited/removed with confirmation, then totals and visible assignments update correctly.

Tasks

- [] Connect candidates and validation to real API responses.
- [] Implement save/edit/remove interactions on the single board.
- [] Keep the three rule results visible during the decision.
- [] Refresh hours and assignments after every successful change.
- [] Add UI/integration tests for each failed rule and valid edit/remove.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Member 5 enforces rules on server; Member 3 sees resulting tutor obligations.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S2-M2-3 | Organiser timesheet and overflow approvals with tests

GitHub issue ID	S2-M2-3	Story points	4
Branch	s2/m2/organiser-approvals	PR reviewer	Member 5
Labels	sprint-2, member-2, area-organiser, points-4	PR base	develop
Depends on	Member 3 timesheets, Member 4 volunteer requests and Member 5 endpoints	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want to approve/return timesheets and approve/reject volunteer claims so that tutor work and overflow assignments are controlled.

Acceptance tests

- [] Given a submitted timesheet, when approved, then its signed-off status is visible to organiser and tutor.
- [] Given a timesheet needing correction, when returned with a reason, then the tutor sees the reason and can edit again.
- [] Given a pending volunteer claim, when approved/rejected, then the student sees the outcome and work availability updates.
- [] Given the wrong role or invalid transition, when attempted, then the action is rejected.

Tasks

- [] Build organiser queues for submitted timesheets and volunteer claims.
- [] Add approve/return/reject actions with required reasons where appropriate.
- [] Show status/history and prevent double actions.
- [] Add transition, wrong-role and failure UI tests.
- [] Run local coverage and resolve critical uncovered states.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Members 3 and 4 consume the resulting statuses; Coordinator receives demo-ready accounts/data.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 2 - Member 3

YOUR SPRINT OWNERSHIP

Tutor dashboard, availability, timesheets, excuses, imports and swaps

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S2-M3-1 | Connect tutor dashboard to real assignments

GitHub issue ID	S2-M3-1	Story points	4
Branch	s2/m3/tutor-dashboard-api	PR reviewer	Member 1
Labels	sprint-2, member-3, area-tutor, points-4	PR base	develop
Depends on	Member 5 own-dashboard endpoint and Member 2 allocations	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a tutor, I want to see only my real assignments, dates, expected work and remaining hours so that I know what I owe.

Acceptance tests

- [] Given the signed-in tutor, when the dashboard loads, then only that tutor's assignments are returned.
- [] Given new/edited allocation, when refreshed, then the dashboard reflects the change.
- [] Given no assignment, slow response or error, when loading, then the correct clear state is shown.
- [] Given another tutor identifier, when attempted, then no other tutor's records are exposed.

Tasks

- [] Replace mock dashboard data with the handwritten API.
- [] Display date/time, course/session, expected hours and status.
- [] Keep loading, empty, error and retry behavior.
- [] Add own-data and cross-user integration tests.
- [] Check phone layout and keyboard access.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Member 2 confirms allocation changes; Member 5 fixes data/permission contract issues.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M3-2 | Work logs and timesheet submission

GitHub issue ID	S2-M3-2	Story points	4
Branch	s2/m3/timesheet-basic	PR reviewer	Member 1
Labels	sprint-2, member-3, area-timesheet, points-4	PR base	develop
Depends on	S2-M3-1 and Member 5 work-log/timesheet endpoints	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a tutor, I want to add worked hours, build a timesheet and submit it so that an organiser can sign it off.

Acceptance tests

- [] Given a valid assigned session and hours, when a work log is saved, then it appears in the draft timesheet.
- [] Given invalid/duplicate/excess hours, when saved, then the request is rejected with a useful message.
- [] Given a draft timesheet, when submitted, then it becomes read-only until approved or returned.
- [] Given organiser approval/return, when the tutor reloads, then the correct status and reason are visible.

Tasks

- [] Connect add/edit/remove work logs to real endpoints.
- [] Build draft totals and submit confirmation.
- [] Lock submitted/approved entries appropriately.
- [] Display returned reason and re-edit path.
- [] Add state-transition and boundary tests.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Member 2 consumes submitted timesheets; Member 6 validates totals/status constraints.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M3-3 | Tutor excuse workflow and Basic regression tests

GitHub issue ID	S2-M3-3	Story points	4
Branch	s2/m3/excuse-regression	PR reviewer	Member 1
Labels	sprint-2, member-3, type-test, points-4	PR base	develop
Depends on	S2-M3-1, S2-M3-2 and Member 5 excuse endpoint	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a tutor, I want to excuse myself from an assigned session and see the result so that the organiser knows I cannot attend.

Acceptance tests

- [] Given an assigned future session and reason, when an excuse is submitted, then it is linked to that assignment and confirmation is shown.
- [] Given a duplicate, past/unassigned session or missing reason, when submitted, then it is rejected clearly.
- [] Given own dashboard/work-log/timesheet/excuse paths, when the regression suite runs, then every Basic tutor journey passes.
- [] Given wrong-user access, when attempted, then no other tutor's data is returned.

Tasks

- [] Connect excuse form to real API and show status.
- [] Prevent duplicate submissions while pending.
- [] Write success, duplicate, invalid-session and permission tests.
- [] Run the full tutor Basic regression locally and in CI.
- [] Fix the highest-priority tutor defects found.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Coordinator receives a complete tutor demo script result; Member 2 sees excuse/availability impact.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 2 - Member 4

YOUR SPRINT OWNERSHIP

Student overflow plus shared responsive and accessible interface

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S2-M4-1 | Connect overflow list to real open work

GitHub issue ID	S2-M4-1	Story points	4
Branch	s2/m4/overflow-api	PR reviewer	Member 6
Labels	sprint-2, member-4, area-student, points-4	PR base	develop
Depends on	Member 5 overflow endpoint and Member 6 schema	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a student, I want to see only real open overflow work so that I do not volunteer for closed or unsuitable items.

Acceptance tests

- [] Given open work, when the page loads, then course, time, hours and status come through the handwritten API.
- [] Given closed/filled/past work, when the list loads, then it is excluded or clearly unavailable.
- [] Given no work, slow response or failure, when loading, then the correct plain state is shown.
- [] Given Student role, when data is returned, then private tutor/organiser fields are not exposed.

Tasks

- [] Replace mock items with the backend endpoint.
- [] Display/filter only permitted open-work fields.
- [] Keep loading, empty, error and retry states.
- [] Add open/closed/past/error integration tests.
- [] Check responsive and keyboard use.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 5 fixes response/permission issues; Member 6 supplies reproducible open/closed seed cases.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M4-2 | Volunteer request and outcome statuses

GitHub issue ID	S2-M4-2	Story points	4
Branch	s2/m4/volunteer-basic	PR reviewer	Member 6
Labels	sprint-2, member-4, area-student, points-4	PR base	develop
Depends on	S2-M4-1 and Member 5 volunteer endpoint	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a student, I want to volunteer once and see pending/accepted/rejected status so that I know what will happen next.

Acceptance tests

- [] Given open work, when confirmed, then one volunteer request is created and shown as pending.
- [] Given a duplicate or already-filled item, when requested, then no second valid claim is created and the reason is shown.
- [] Given organiser approval/rejection, when reloaded, then the student sees the correct outcome.
- [] Given a request in flight, when clicked repeatedly, then only one call is made.

Tasks

- [] Connect confirmation to real volunteer endpoint.
- [] Display pending/accepted/rejected statuses.
- [] Prevent duplicate clicks and handle conflict responses.
- [] Add tests for success, duplicate, race/conflict and outcome states.
- [] Generate coverage and fix uncovered critical branches.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 2 consumes pending claims; Member 5/6 enforce uniqueness and atomic outcome.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M4-3 | External-result UI, accessibility and student regression

GitHub issue ID	S2-M4-3	Story points	4
Branch	s2/m4/shared-quality	PR reviewer	Member 6
Labels	sprint-2, member-4, type-test, points-4	PR base	develop
Depends on	Member 6 external adapter and all Basic shared UI	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As any user, I want the external information and shared UI to remain understandable when services are slow or unavailable so that core Tutor Mx work still succeeds.

Acceptance tests

- [] Given a successful external response, when its relevant screen opens, then the useful result is shown with source/time context.
- [] Given timeout/unavailable/invalid response, when displayed, then fallback wording appears and core actions remain usable.
- [] Given keyboard-only use and common widths, when Basic role journeys run, then primary actions remain reachable and labelled.
- [] Given the student regression suite, when run, then list/volunteer/status/error paths pass in CI.

Tasks

- [] Connect the safe backend external-result endpoint to its agreed UI location.
- [] Add loading, unavailable, retry/fallback behavior.
- [] Run keyboard/focus/label and common-width checks across shared/student UI.
- [] Automate critical external success/failure and student cases.
- [] Fix assigned accessibility or responsive failures.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Coordinator receives accessibility/test evidence; Member 6 receives any adapter failure cases.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 2 - Member 5

YOUR SPRINT OWNERSHIP

Separate handwritten backend HTTP API, rules, security and API tests

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S2-M5-1 | Complete Basic organiser and allocation API

GitHub issue ID	S2-M5-1	Story points	4
Branch	s2/m5/organiser-api	PR reviewer	Member 2
Labels	sprint-2, member-5, area-api, points-4	PR base	develop
Depends on	Member 6 Basic schema and Member 1 role matrix	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As the organiser UI, I want complete protected course, tutor and allocation endpoints so that the Basic organiser journey works through the handwritten API.

Acceptance tests

- [] Given valid organiser requests, when course/tutor/allocation create/read/update/delete endpoints run, then data persists correctly.
- [] Given failed mark/clash/hour rule, when allocation save is requested, then the server rejects it with stable reason data.
- [] Given invalid input, missing records or wrong role, when called, then consistent 400/404/403 responses are returned.
- [] Given direct API use, when the frontend is bypassed, then all rules and permissions are still enforced.

Tasks

- [] Complete organiser course/tutor CRUD endpoints.
- [] Complete candidate validation and allocation CRUD endpoints.
- [] Recalculate all three rules inside the save transaction.
- [] Add endpoint tests for success, boundaries, validation and permissions.
- [] Publish a deployed health/API URL for team testing.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Member 2 connects the board; Member 6 tunes queries/constraints.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M5-2 | Complete Basic tutor and student API

GitHub issue ID	S2-M5-2	Story points	4
Branch	s2/m5/tutor-student-api	PR reviewer	Member 2
Labels	sprint-2, member-5, area-api, points-4	PR base	develop
Depends on	Member 6 Basic workflow schema and Member 1 ownership rules	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As tutor and student interfaces, we want complete own-data, timesheet, excuse, overflow and volunteer endpoints so that every Basic journey uses the same backend.

Acceptance tests

- [] Given a tutor, when dashboard/work-log/timesheet/excuse endpoints run, then only their permitted records are read/changed.
- [] Given a student, when overflow/volunteer endpoints run, then only safe open-work data is returned and duplicates are rejected.
- [] Given organiser approvals, when valid transitions run, then tutors/students see the resulting state.
- [] Given invalid transitions or cross-user identifiers, when called, then the server rejects them.

Tasks

- [] Complete own-dashboard, work-log, timesheet and excuse endpoints.
- [] Complete overflow list, volunteer and organiser approval endpoints.
- [] Implement ownership and state-transition checks.
- [] Add end-to-end API tests for all Basic role journeys.
- [] Check logs/errors do not expose tokens or personal fields.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Members 3 and 4 connect their screens; Member 6 enforces database uniqueness/transitions.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M5-3 | Standard errors, external deployment and API coverage

GitHub issue ID	S2-M5-3	Story points	4
Branch	s2/m5/api-quality	PR reviewer	Member 2
Labels	sprint-2, member-5, type-test, points-4	PR base	develop
Depends on	S2-M5-1, S2-M5-2 and CI from Member 6	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want a stable externally available API with consistent errors and useful tests so that frontend work and Sprint 2 review are reliable.

Acceptance tests

- [] Given every Basic endpoint, when invalid input/auth/not-found/server failure occurs, then the agreed JSON error shape is used.
- [] Given the deployed API, when /health and representative protected calls run, then it is externally reachable through HTTPS.
- [] Given CI, when API tests run, then success, boundary, permission and failure cases produce coverage uploaded to Codecov.
- [] Given a deployment failure, when it occurs, then CI shows the failed step and the previous working release is not silently replaced.

Tasks

- [] Apply central validation/error handling to all Basic endpoints.
- [] Deploy the backend and configure safe environment secrets.
- [] Add smoke tests for health and representative API calls.
- [] Ensure API coverage file uploads with the backend Codecov flag.
- [] Fix high-priority API defects from other Members.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Integration Lead uses health/tests/Codecov as merge evidence; Coordinator uses route examples for API docs.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 2 - Member 6

YOUR SPRINT OWNERSHIP

Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery
3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S2-M6-1 | Complete Basic Neon PostgreSQL schema and constraints

GitHub issue ID	S2-M6-1	Story points	4
Branch	s2/m6/basic-workflow-schema	PR reviewer	Member 4
Labels	sprint-2, member-6, area-database, points-4	PR base	develop
Depends on	Final Basic payloads from Members 2-5	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As the team, we want the database to support every Basic journey and reject invalid relationships so that application state remains trustworthy.

Acceptance tests

- [] Given clean migrations, when run, then work logs, timesheets, excuses, overflow work and volunteer claims are created with required relations.
- [] Given invalid status transition, orphan record, duplicate claim or invalid hours, when inserted, then a constraint/transaction rejects it.
- [] Given safe seed data, when loaded, then all three roles and Basic pass/fail demo cases are reproducible.
- [] Given reset/redeploy, when run, then no manual production schema editing is required.

Tasks

- [] Add workflow tables/enums/status constraints via migrations.
- [] Add indexes for common dashboard/board/open-work queries.
- [] Create realistic non-personal seed cases.
- [] Add migration, constraint and repeatability tests.
- [] Verify secrets and production data are absent from repository.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Member 5 connects transactions/queries; Members 2-4 use stable demo data.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M6-2 | Production external API adapter with graceful fallback

GitHub issue ID	S2-M6-2	Story points	4
Branch	s2/m6/external-api-adapter	PR reviewer	Member 4
Labels	sprint-2, member-6, area-integration, points-4	PR base	develop
Depends on	Approved Sprint 1 external service and Member 5 backend	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a Tutor Mx user, I want the relevant external information delivered through the backend so that credentials stay safe and outages do not block core work.

Acceptance tests

- [] Given valid service response, when the handwritten endpoint is called, then the app receives only the team's validated safe format.
- [] Given timeout, rate limit, unavailable service or invalid payload, when called, then a controlled fallback/status is returned.
- [] Given source credentials, when repository/browser/logs are inspected, then secrets are absent.
- [] Given mocked service states, when tests run, then success and every agreed failure path pass.

Tasks

- [] Implement the adapter in backend-side code.
- [] Add timeout, response schema validation and safe error mapping.
- [] Add caching only if justified; never hide stale/source time.
- [] Add mock tests for success, timeout, rate limit and invalid payload.
- [] Coordinate the route with Member 5 and UI result with Member 4.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Member 5 exposes the adapter through handwritten API; Member 4 renders success/fallback.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S2-M6-3 | Allocation queries, deployment smoke and Codecov reliability

GitHub issue ID	S2-M6-3	Story points	4
Branch	s2/m6/data-delivery-quality	PR reviewer	Member 4
Labels	sprint-2, member-6, type-test, points-4	PR base	develop
Depends on	S2-M6-1, Member 5 API and GitHub Actions	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the Integration Lead, I want efficient reproducible queries and reliable CI/deployment checks so that the complete Basic system can be tested before merge.

Acceptance tests

- [] Given a course/session, when candidate queries run, then marks, clashes, used/remaining hours and open work are correct.
- [] Given clean migration plus seed, when the API smoke test runs, then all three roles complete their core read path.
- [] Given frontend/backend coverage, when a PR runs, then both reports appear under separate Codecov flags.
- [] Given a broken migration/test/health check, when pushed, then CI fails and merge is blocked.

Tasks

- [] Implement/test candidate, hour and open-work queries.
- [] Add clean-database setup and deployed health smoke jobs.
- [] Verify frontend/backend Codecov flags and required checks.
- [] Troubleshoot flaky or path-dependent workflow steps.
- [] Run a clean end-to-end CI rehearsal on develop.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Integration Lead uses this as Sprint 2 merge gate; Coordinator receives run links and screenshots.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 2: integration review and official rubric

OFFICIAL DESCRIPTION NOTE

Weights and official level meanings come from project_briefs.pdf. The evidence wording below is condensed into Tutor Mx language for planning; use the official PDF when marks are disputed.

Integration Lead release gate

- [] Every issue is linked to a reviewed PR targeting develop; no Member pushes directly to main.
- [] The assigned reviewer checks the issue's acceptance tests, changed code, local/CI tests, coverage and AI attribution.
- [] Integration Lead merges approved PRs into develop one at a time, resolves conflicts with the author, then runs the full suite.
- [] Only the Integration Lead opens develop -> main. Main remains protected and merges only after required checks pass.
- [] The Project Coordinator runs meetings, maintains the GitHub board/decisions and writes project documentation from Member evidence.

Sprint review demonstration checklist

- [] Run complete Basic Organiser allocation and approval paths.
- [] Run complete Tutor dashboard, work-log, timesheet and excuse paths.
- [] Run complete Student overflow volunteer and organiser outcome path.
- [] Trigger all three allocation failures and external API fallback.
- [] Open UI/API tests and Codecov frontend/backend results.

Official rubric weights

Core Features	25%	Basic core is implemented with at most minor defects.
Automated Testing	10%	Useful UI and API testing, not only default tests.
Stakeholder Reviews	10%	Regular feedback is evaluated and integrated.
API	15%	Handwritten API is externally available/documented and external API is integrated.
User Feedback	10%	Formal representative-user process with evidence and integrated findings.
Project Methodology	10%	Sprint method is actively followed.
Bug Tracker	5%	Bugs are recorded and used continuously.
Database Documentation	5%	Schema, deployment and motivation are current.
Third-Party Code Documentation	5%	Third-party code is identified and motivated.
Testing Documentation	5%	Automated/user testing process and policy are documented.
TOTAL	100%	Official weights total 100%.

Sprint 3: plan, order and handovers

SPRINT GOAL

16-29 September 2026

Implement Intermediate scope, user-feedback fixes and near-complete quality

What this sprint must achieve

- Complete Intermediate features: ranking, bulk actions, shortages, overrides, imports, timesheet workflow, notifications, reports, search and onboarding.
- Apply the Coordinator's feedback priorities and expand regression, security and performance tests.
- Finish with a near-complete product and no hidden Basic regression.

Equal technical allocation

MEMBER	TECHNICAL OWNERSHIP	ISSUES	POINTS
Member 1	Authentication, role security, notifications/search and shared release checks	S3-M1-1 to S3-M1-3	12
Member 2	Organiser interface, allocation board, approvals, reports and scenarios	S3-M2-1 to S3-M2-3	12
Member 3	Tutor dashboard, availability, timesheets, excuses, imports and swaps	S3-M3-1 to S3-M3-3	12
Member 4	Student overflow plus shared responsive and accessible interface	S3-M4-1 to S3-M4-3	12
Member 5	Handwritten backend HTTP API, validation, business rules, security and API tests	S3-M5-1 to S3-M5-3	12
Member 6	Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery	-1 to S3-M6-3	12
TOTAL	Meetings and documentation are owned by the Project Coordinator.	18 issues	72

BEGINNER RULE

A contract is a written example of the data, API request, response and error shape. Once the example is agreed, frontend work may begin with a mock. A member waits only for the exact missing contract - not for the whole backend or whole sprint.

Sprint 3: who goes first and how work joins

WHO STARTS FIRST

Member 6 starts with Intermediate data, allocation-engine and background-job contracts. Member 5 begins each endpoint when its contract is usable. Members 1-4 start their feature screens and tests immediately with approved mocks, rather than waiting for the whole backend.

Detailed order, parallel work and handover

STEP / WHEN	OWNER	DO THIS	SAFE PARALLEL WORK	HAND OVER TO	READY WHEN
1 Day 1	Member 6	Publish Intermediate schema changes, ranking inputs/outputs, job states and migration plan.	M1-4 break features into UI states and tests; M5 prepares route skeletons.	Member 5 + Members 1-4	Contracts include success, validation and failure examples.
2 Per ready contract	Member 5	Implement Intermediate routes, server-side permissions and business rules in reviewable groups.	M6 tunes queries/jobs. M1 tests search/notification permissions.	Relevant UI owner	Route tests pass and payload examples are posted before integration.
3 Same time	Members 1-4	Build role features with exact mocks: organiser tools, tutor workflows, student overflow, search/onboarding and shared states.	Each member works in their owned area; shared components have one named owner.	Member 5	UI/component tests pass and contract questions are GitHub blockers.
4 As endpoint groups land	Relevant UI owner	Replace the matching mock, test the real journey and report API defects with request/response evidence.	Other owners keep building against stable mocks; M5 fixes shared defects.	Member 6	Connected journey passes without breaking the Basic flow.
5 Feature-complete	All members	Apply Coordinator-prioritised feedback and add regression tests for every accepted fix.	Independent fixes run in parallel; overlapping changes are coordinated by the file/contract owner.	Issue reviewers	Each fix reproduces the old failure and proves the corrected behavior.
6 Quality gate	Members 1, 5 + 6	Run permission/security, API, migration, performance, external-failure, CI and Codecov checks.	Members 2-4 run role regression, keyboard and responsive checks.	Integration Lead	No Basic regression and no unowned severity-one/two defect remains.
7 Finish	Integration Lead	Merge reviewed PRs to develop one at a time and run the near-complete release rehearsal.	Coordinator collects performance, feedback and test evidence.	Project Coordinator	Near-complete build and honest remaining-risk list are ready for review.

How to hand over: post the contract or PR link in the dependent GitHub issue; include setup steps, sample data/request/response, tests run and known limits; tag the named receiver. The receiver replies that they can use it or opens a blocking issue naming the missing item. A handover is not complete just because code was pushed.

Sprint 3 - Member 1

YOUR SPRINT OWNERSHIP

Authentication, role security, notifications/search and shared release checks

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S3-M1-1 | Notification inbox and due reminders

GitHub issue ID	S3-M1-1	Story points	4
Branch	s3/m1/notifications	PR reviewer	Member 4
Labels	sprint-3, member-1, area-notifications, points-4	PR base	develop
Depends on	Member 6 notification data/job and Member 5 endpoints	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As an affected user, I want assignment changes and due items to reach me so that I do not miss important Tutor Mx work.

Acceptance tests

- [] Given an assignment/status change affecting a user, when it occurs, then one relevant notification appears for that user.
- [] Given a due timesheet/approval, when the reminder rule runs, then the person currently responsible is notified.
- [] Given an unrelated user, when notifications load, then they cannot see another user's messages.
- [] Given a notification link, when opened, then it leads to the permitted relevant item or a safe unavailable state.

Tasks

- [] Build notification inbox/list and unread/read state.
- [] Add deep links for permitted assignment/timesheet/claim items.
- [] Connect to Member 5 endpoints and Member 6 reminder data.
- [] Add own-data, duplicate and missing-target tests.
- [] Run coverage and fix critical notification branches.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Members 2-4 reuse notification components; Member 5/6 receive duplication/permission failures.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S3-M1-2 | Global search and guided tutor onboarding

GitHub issue ID	S3-M1-2	Story points	4
Branch	s3/m1/search-onboarding	PR reviewer	Member 4
Labels	sprint-3, member-1, area-usability, points-4	PR base	develop
Depends on	Member 5 search endpoint and role navigation	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a user, I want fast search and first-time guidance so that I can find a course, person or timesheet without learning the whole system.

Acceptance tests

- [] Given an authorised query, when search runs, then permitted courses/people/timesheets are returned with type labels.
- [] Given a short/empty/no-result query, when searched, then the response is clear and does not leak hidden data.
- [] Given a first-time tutor, when onboarding starts, then it guides account/profile, availability and dashboard steps and can be skipped/resumed.
- [] Given keyboard-only use, when search/onboarding runs, then focus and selection are usable.

Tasks

- [] Build shared search control and grouped results.
- [] Apply role-safe result rendering and no-result/error states.
- [] Build tutor onboarding steps with progress and skip/resume.
- [] Add permission, keyboard and state tests.
- [] Check common phone/desktop layouts.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Member 5 supplies role-filtered search; Member 3 links onboarding to tutor screens.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S3-M1-3 | Shared performance, accessibility and feedback fixes

GitHub issue ID	S3-M1-3	Story points	4
Branch	s3/m1/shared-quality-fixes	PR reviewer	Member 4
Labels	sprint-3, member-1, type-test, points-4	PR base	develop
Depends on	Coordinator's prioritised user-feedback issues and integrated Sprint 3 app	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want high-priority shared problems fixed and measured so that the near-complete product is usable, secure and fast enough.

Acceptance tests

- [] Given prioritised feedback assigned to Member 1, when fixed, then the linked acceptance test passes and the issue references the PR.
- [] Given account/search/notification critical paths, when measured, then they meet the team's recorded response target or a justified limitation is logged.
- [] Given keyboard/focus/labels, when shared paths are checked, then no critical failure remains.
- [] Given regression tests, when CI runs, then previous Basic journeys still pass.

Tasks

- [] Implement the highest-priority shared feedback fixes assigned by the Coordinator.
- [] Measure and improve one slow shared path.
- [] Run focused accessibility checks and fix critical findings.
- [] Expand regression tests for account/search/notifications.
- [] Attach before/after evidence and coverage to PR.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Coordinator closes feedback items only after evidence; Integration Lead checks regression/Codecov.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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Sprint 3 - Member 2

YOUR SPRINT OWNERSHIP

Organiser interface, allocation board, approvals, reports and scenarios

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S3-M2-1 | Ranked candidates and shortage visibility

GitHub issue ID	S3-M2-1	Story points	4
Branch	s3/m2/ranked-candidates	PR reviewer	Member 6
Labels	sprint-3, member-2, area-allocation, points-4	PR base	develop
Depends on	Member 5 ranking endpoint and Member 6 ranking/query logic	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As an organiser, I want eligible tutors ranked and shortages visible so that the system does more of the allocation legwork.

Acceptance tests

- [] Given a course/session, when candidates load, then eligible tutors are ordered by the documented ranking inputs.
- [] Given an ineligible tutor, when listed/excluded, then the organiser can see the reason.
- [] Given unmet required work, when the board renders, then the shortage amount/location is visible.
- [] Given equal scores or missing data, when ranking occurs, then the tie/unknown behavior is deterministic and explained.

Tasks

- [] Connect ranked candidate results to the board.
- [] Show concise reasons/inputs without pretending the score is objective truth.
- [] Add shortage indicators at course/session level.
- [] Add tests for ranking order, ties, excluded tutors and missing data.
- [] Keep all three Basic rule results visible.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 6 owns algorithm/data correctness; Member 5 owns response stability.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S3-M2-2 | Bulk fill and reasoned organiser override

GitHub issue ID	S3-M2-2	Story points	4
Branch	s3/m2/bulk-override	PR reviewer	Member 6
Labels	sprint-3, member-2, area-allocation, points-4	PR base	develop
Depends on	S3-M2-1 and Member 5 bulk/override endpoints	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want obvious assignments filled in bulk and exceptions overridden with a recorded reason so that repetitive allocation is faster but accountable.

Acceptance tests

- [] Given safe obvious assignments, when previewed in bulk, then proposed changes are listed before confirmation.
- [] Given confirmation, when bulk fill succeeds/fails partly, then each item has a clear result and no hidden partial state.
- [] Given an allowed override, when submitted, then a non-empty reason is required and shown in history.
- [] Given an override that must never be allowed, when attempted, then the backend rejects it.

Tasks

- [] Build bulk preview, select/deselect and confirm UI.
- [] Handle full success, partial result and failure states.
- [] Build override action with required reason and warning.
- [] Add tests for preview/confirm/cancel/partial/forbidden cases.
- [] Refresh board totals and shortage after changes.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 5/6 define transaction and allowed-override rules; Coordinator receives audit evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S3-M2-3 | Hours, budget and distribution reports with keyboard board use

GitHub issue ID	S3-M2-3	Story points	4
Branch	s3/m2/organiser-reports	PR reviewer	Member 6
Labels	sprint-3, member-2, type-test, points-4	PR base	develop
Depends on	Member 5 report endpoints and Coordinator-approved budget fields	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want hours, spend and work distribution plus efficient keyboard actions so that I can monitor and operate a large allocation board.

Acceptance tests

- [] Given allocations/timesheets, when reports load, then hours per course, spend against budget and work distribution use documented calculations.
- [] Given no/partial data, when reports load, then totals state the limitation rather than inventing values.
- [] Given keyboard-only use, when common board actions run, then selection, validation and bulk actions can be completed.
- [] Given large realistic seed data, when board/report tests run, then measured response remains within the team target.

Tasks

- [] Build organiser report views/summary cards with clear units.
- [] Connect to backend calculations; do not recalculate authoritative totals differently in UI.
- [] Add keyboard shortcuts/actions with visible help and no conflict with browser basics.
- [] Add calculation-display, keyboard and large-data tests.
- [] Implement assigned performance fix if target is missed.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 5/6 provide calculations/load data; Coordinator records formula/target in docs.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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Sprint 3 - Member 3

YOUR SPRINT OWNERSHIP

Tutor dashboard, availability, timesheets, excuses, imports and swaps

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S3-M3-1 | Timetable import, recurring exceptions and term calendar

GitHub issue ID	S3-M3-1	Story points	4
Branch	s3/m3/timetable-import	PR reviewer	Member 5
Labels	sprint-3, member-3, area-tutor, points-4	PR base	develop
Depends on	Member 5 import/availability endpoints and Member 6 calendar schema	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a tutor, I want to import my timetable and add recurring exceptions so that availability is easier to provide accurately.

Acceptance tests

- [] Given a supported timetable file/format, when imported, then a preview shows parsed busy times before confirmation.
- [] Given invalid/partial input, when imported, then errors identify the affected rows without corrupting existing availability.
- [] Given recurring exception and term dates, when saved, then generated busy occurrences fall only within the term rule.
- [] Given imported availability, when an allocation is checked, then clashes use the confirmed data.

Tasks

- [] Build upload/paste, preview, error and confirm UI.
- [] Add recurring-exception and term-calendar controls.
- [] Connect import/availability endpoints.
- [] Add valid, invalid, partial and boundary-date tests.
- [] Check keyboard/focus and narrow layout.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Member 6 owns parser data/term rules; Member 2 sees resulting allocation clashes.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S3-M3-2 | Timesheet corrections, disputes and payroll export

GitHub issue ID	S3-M3-2	Story points	4
Branch	s3/m3/timesheet-workflow	PR reviewer	Member 5
Labels	sprint-3, member-3, area-timesheet, points-4	PR base	develop
Depends on	Member 5 workflow/export endpoints and Member 6 status history	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a tutor and organiser, we want corrections/disputes and payroll-ready output so that timesheets move through a complete accountable workflow.

Acceptance tests

- [] Given a returned timesheet, when corrected/resubmitted, then previous values/reason remain traceable.
- [] Given a disputed decision, when a dispute is submitted, then status and reason route to the responsible organiser.
- [] Given an approved period, when export is requested, then the file contains the agreed fields/totals and excludes unapproved work.
- [] Given invalid transition or unauthorised user, when attempted, then it is rejected.

Tasks

- [] Build tutor correction and dispute views.
- [] Show status timeline/history clearly.
- [] Connect payroll export action and safe download state.
- [] Add transition, permission, totals and export-content tests.
- [] Fix any duplicate/rounding/date boundary defects found.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Member 2 handles organiser side; Coordinator receives sample anonymised export evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S3-M3-3 | Tutor onboarding/notification integration and regression

GitHub issue ID	S3-M3-3	Story points	4
Branch	s3/m3/tutor-intermediate-tests	PR reviewer	Member 5
Labels	sprint-3, member-3, type-test, points-4	PR base	develop
Depends on	S3-M1-1, S3-M1-2, S3-M3-1 and S3-M3-2	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a tutor, I want onboarding and notifications connected to my real tasks so that the expanded workflow remains understandable.

Acceptance tests

- [] Given a first-time tutor, when onboarding completes, then profile/availability/dashboard steps point to working screens.
- [] Given timesheet, excuse or allocation status change, when notification opens, then it leads to the correct permitted item.
- [] Given the Intermediate tutor regression, when run, then Basic plus import/exceptions/corrections/disputes/export pass.
- [] Given user-feedback fixes assigned to tutor area, when re-tested, then the original failed step now passes.

Tasks

- [] Connect onboarding steps and notification links to tutor routes.
- [] Write Intermediate tutor end-to-end/regression tests.
- [] Implement assigned high-priority tutor feedback fixes.
- [] Run coverage and add tests for uncovered transitions.
- [] Run common-width and keyboard checks.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Coordinator closes feedback issues; Integration Lead uses tutor regression as merge gate.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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Sprint 3 - Member 4

YOUR SPRINT OWNERSHIP

Student overflow plus shared responsive and accessible interface

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S3-M4-1 | Overflow posting, claim and organiser-approval UI

GitHub issue ID	S3-M4-1	Story points	4
Branch	s3/m4/overflow-workflow	PR reviewer	Member 1
Labels	sprint-3, member-4, area-overflow, points-4	PR base	develop
Depends on	Member 5 overflow workflow API and Member 6 schema	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As organiser and student, we want overflow work posted, claimed and approved so that unassigned work reaches volunteers with control.

Acceptance tests

- [] Given an organiser, when valid overflow work is posted/closed, then student availability updates.
- [] Given a student, when an open item is claimed, then one pending claim is created.
- [] Given organiser approval/rejection, when decided, then only the approved outcome changes the assignment and the student is notified.
- [] Given two near-simultaneous claims, when processed, then the UI shows the authoritative result without two approvals.

Tasks

- [] Build organiser post/edit/close overflow UI.
- [] Complete student claim status/history UI.
- [] Build organiser claim approval/rejection UI or integrate Member 2 component.
- [] Add open/close/duplicate/race/outcome tests.
- [] Connect notifications and refresh affected views.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Member 5/6 enforce atomic claim rules; Member 1 supplies notification component.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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END OF COPY

S3-M4-2 | Shared search/notification states and bulk-action consistency

GitHub issue ID	S3-M4-2	Story points	4
Branch	s3/m4/shared-intermediate-ui	PR reviewer	Member 1
Labels	sprint-3, member-4, area-shared-ui, points-4	PR base	develop
Depends on	Member 1 search/notifications and Member 2 bulk UI	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As any user, I want new search, notification and bulk interactions to use consistent accessible states so that the larger system still feels predictable.

Acceptance tests

- [] Given loading/empty/error/success for new features, when shown, then shared components use consistent text, focus and announcements.
- [] Given unread notifications/search results, when keyboard navigation is used, then selection and return focus behave correctly.
- [] Given bulk confirmation, when cancel/partial failure occurs, then the final state is clear.
- [] Given common screen widths, when shared Intermediate features render, then primary actions are not clipped.

Tasks

- [] Extend shared status/confirmation components for new Intermediate cases.
- [] Integrate accessible live-region/focus behavior where needed.
- [] Review and fix search, notification and bulk responsive issues.
- [] Add shared component and keyboard tests.
- [] Remove inconsistent duplicate UI patterns.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Members 1 and 2 adopt shared fixes; Coordinator records accessibility evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S3-M4-3 | Representative-user fixes and app-wide accessibility regression

GitHub issue ID	S3-M4-3	Story points	4
Branch	s3/m4/usability-regression	PR reviewer	Member 1
Labels	sprint-3, member-4, type-test, points-4	PR base	develop
Depends on	Coordinator's prioritised user-test issues and near-complete develop build	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want representative-user problems and app-wide accessibility failures fixed so that Organiser, Tutor and Student journeys are usable.

Acceptance tests

- [] Given each representative role script, when run on the shared build, then critical tasks can be completed without coaching.
- [] Given assigned feedback findings, when fixed/re-tested, then evidence links issue, PR and passing step.
- [] Given keyboard/focus/label/status checks, when app-wide regression runs, then no critical failure remains.
- [] Given phone/tablet/desktop agreed widths, when core role journeys run, then no primary action is clipped.

Tasks

- [] Implement high-priority user-feedback/accessibility fixes assigned to Member 4.
- [] Run a repeatable app-wide keyboard/focus/label checklist.
- [] Run responsive checks across the three role journeys.
- [] Automate critical shared failures where practical.
- [] Attach re-test and coverage evidence to PR.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Coordinator owns sessions/notes; Member 4 implements technical fixes only.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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Sprint 3 - Member 5

YOUR SPRINT OWNERSHIP

Separate handwritten backend HTTP API, rules, security and API tests

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S3-M5-1 | Ranking, bulk-fill and override API

GitHub issue ID	S3-M5-1	Story points	4
Branch	s3/m5/allocation-intermediate-api	PR reviewer	Member 3
Labels	sprint-3, member-5, area-api, points-4	PR base	develop
Depends on	Member 6 ranking logic/schema and Basic allocation API	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As the organiser UI, I want protected ranking, shortage, bulk and override endpoints so that Intermediate allocation behavior is consistent and auditable.

Acceptance tests

- [] Given a course/session, when ranking endpoint runs, then deterministic ordered candidates and reasons are returned.
- [] Given bulk preview/commit, when requested, then preview has no side effects and commit returns per-item results atomically or explicitly partially.
- [] Given allowed override, when a reason is supplied, then it is stored; missing reason is rejected.
- [] Given forbidden override/wrong role, when attempted, then no allocation changes.

Tasks

- [] Implement ranking/shortage endpoints using Member 6 services.
- [] Implement bulk preview and commit contract.
- [] Implement allowed override with required reason/history.
- [] Add deterministic, partial/failure, permission and audit tests.
- [] Update API examples as raw handoff to Coordinator.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Member 2 consumes endpoints; Member 6 validates algorithm/transaction behavior.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S3-M5-2 | Intermediate workflow, search, report and notification API

GitHub issue ID	S3-M5-2	Story points	4
Branch	s3/m5/intermediate-api	PR reviewer	Member 3
Labels	sprint-3, member-5, area-api, points-4	PR base	develop
Depends on	Member 6 Intermediate schema/jobs	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the expanded frontend, I want protected endpoints for timetable, timesheet workflow, overflow, search, reports and notifications so that Intermediate journeys stay behind the handwritten API.

Acceptance tests

- [] Given supported timetable/correction/dispute/overflow requests, when valid, then state transitions persist with history.
- [] Given search, when called, then results are role-filtered and paginated/limited.
- [] Given reports, when called, then hours/spend/distribution calculations are server-defined and testable.
- [] Given notifications, when read/marked, then only the owning user's records change.

Tasks

- [] Implement/complete timetable and availability endpoints.
- [] Implement correction/dispute/export and overflow workflow endpoints.
- [] Implement role-safe search/report/notification endpoints.
- [] Add validation, permission, state and calculation tests.
- [] Check errors use the shared format.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Members 1-4 connect screens; Member 6 tunes queries/data histories.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S3-M5-3 | API performance, security and Intermediate regression

GitHub issue ID	S3-M5-3	Story points	4
Branch	s3/m5/api-intermediate-quality	PR reviewer	Member 3
Labels	sprint-3, member-5, type-test, points-4	PR base	develop
Depends on	All S3 API work on develop and realistic Member 6 data	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want measured and regression-tested API behavior so that the near-complete system is stable and secure.

Acceptance tests

- [] Given realistic data, when critical board/search/report endpoints are measured, then they meet the recorded target or an issue/limit is documented.
- [] Given malformed/large/unauthorised requests, when sent, then validation, limits and permissions protect the service.
- [] Given full API regression, when CI runs, then all Basic and Intermediate endpoint suites pass and upload coverage.
- [] Given an external dependency failure, when it occurs, then core endpoints degrade gracefully.

Tasks

- [] Add timing/load checks for critical API routes.
- [] Fix obvious unbounded/N+1/validation issues.
- [] Add security regression for roles, ownership, input sizes and logging.
- [] Run full API suite with Codecov backend flag.
- [] Implement highest-priority API feedback/performance fixes.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Integration Lead requires these checks; Coordinator records measured target/results.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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Sprint 3 - Member 6

YOUR SPRINT OWNERSHIP

Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery
3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S3-M6-1 | Intermediate schema, status history and report data

GitHub issue ID	S3-M6-1	Story points	4
Branch	s3/m6/intermediate-schema	PR reviewer	Member 2
Labels	sprint-3, member-6, area-database, points-4	PR base	develop
Depends on	S3 feature payloads from Members 1-5	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As the team, we want migrated data structures for Intermediate workflows so that timetable, corrections, disputes, claims, notifications and reports remain consistent.

Acceptance tests

- [] Given clean migrations, when run, then term calendar/import, recurring exception, status history, notification and budget/report fields exist.
- [] Given invalid transition/duplicate event, when stored, then constraints or transaction logic reject it.
- [] Given report queries, when run on known seed data, then hours/spend/distribution totals match expected fixtures.
- [] Given migration from Sprint 2 data, when applied, then existing Basic records remain valid.

Tasks

- [] Add forward migrations for Intermediate entities/history.
- [] Add constraints/indexes for state, search and reporting queries.
- [] Create known fixtures for ranking/reports/notifications.
- [] Test clean and upgrade migration paths.
- [] Test totals/history/uniqueness boundaries.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Member 5 implements endpoints; Members 1-4 receive stable fixture cases.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S3-M6-2 | Ranking service, reminder jobs and allocation data engine

GitHub issue ID	S3-M6-2	Story points	4
Branch	s3/m6/intermediate-engine	PR reviewer	Member 2
Labels	sprint-3, member-6, area-allocation, points-4	PR base	develop
Depends on	S3-M6-1 and Coordinator-approved ranking inputs	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want deterministic ranking, shortage calculation and reminders so that recommendations and due work are useful and explainable.

Acceptance tests

- [] Given candidate inputs, when ranking runs, then the order and contributing factors are deterministic and testable.
- [] Given shortages, when required versus assigned work is calculated, then the exact gap is returned.
- [] Given a due item, when reminder job runs repeatedly, then duplicate notifications are controlled.
- [] Given missing/invalid data, when engine runs, then it returns an explainable limitation rather than a misleading score.

Tasks

- [] Implement ranking as a pure/testable service with named factors.
- [] Implement shortage and report calculation helpers.
- [] Implement reminder scheduling/job logic with idempotency.
- [] Add tests for ties, missing data, boundaries and repeat runs.
- [] Measure realistic data performance and add needed indexes.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Member 5 exposes services; Members 1/2 render reasons/results.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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S3-M6-3 | Delivery resilience, data performance and Codecov gate

GitHub issue ID	S3-M6-3	Story points	4
Branch	s3/m6/release-quality	PR reviewer	Member 2
Labels	sprint-3, member-6, type-test, points-4	PR base	develop
Depends on	All Intermediate integrations and CI	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the Integration Lead, I want resilient deployment, database/integration tests and meaningful Codecov gates so that Sprint 3 is near-complete rather than only feature-rich.

Acceptance tests

- [] Given slow/unavailable external service, when production-like tests run, then core Tutor Mx journeys remain available.
- [] Given realistic database size, when critical queries run, then measured times meet the recorded target after indexing/tuning.
- [] Given a PR, when coverage changes, then Codecov project/patch checks follow the team's stated policy.
- [] Given failed migration/smoke/coverage check, when merge is attempted, then protection blocks it.

Tasks

- [] Add integration failure/timeout regression and fallback checks.
- [] Measure/tune critical database queries and record results for Coordinator.
- [] Set Codecov flags/status policy (team target, not official mark threshold).
- [] Run clean migration, full CI and deployed smoke rehearsal.
- [] Fix flaky workflows and unsafe secret output.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Integration Lead uses this as Sprint 3 release gate; Coordinator writes methodology/testing evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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Sprint 3: integration review and official rubric

OFFICIAL DESCRIPTION NOTE

The latest project brief publishes the Sprint 3 and final criterion names/weights but leaves Advanced/Intermediate/Basic description cells blank. The evidence wording below is team readiness guidance, not a replacement for the lecturer's rubric.

Integration Lead release gate

- [] Every issue is linked to a reviewed PR targeting develop; no Member pushes directly to main.
- [] The assigned reviewer checks the issue's acceptance tests, changed code, local/CI tests, coverage and AI attribution.
- [] Integration Lead merges approved PRs into develop one at a time, resolves conflicts with the author, then runs the full suite.
- [] Only the Integration Lead opens develop -> main. Main remains protected and merges only after required checks pass.
- [] The Project Coordinator runs meetings, maintains the GitHub board/decisions and writes project documentation from Member evidence.

Sprint review demonstration checklist

- [] Demonstrate ranking, shortages, bulk preview, override reason and reports.
- [] Demonstrate timetable import/exceptions, correction/dispute/export.
- [] Demonstrate overflow approval, notifications, search and onboarding.
- [] Show representative-user issue -> fix -> passing re-test trace.
- [] Show measured performance and full Basic/Intermediate regression.

Official rubric weights

User Feedback	10%	Representative findings are evaluated, prioritised and fixed where appropriate.
Automated Testing	10%	Useful regression/UI/API testing with Codecov evidence.
Feature Implementation	20%	Most planned/claimed features work in the integrated app.
API Implementation	20%	Handwritten API supports integrated claimed features.
Performance	5%	Critical paths are measured against recorded targets.
Improvement	5%	Evidence shows feedback/defect/performance improvements.
Documentation	15%	Public docs match the near-complete product.
Project Methodology	15%	Consistent sprint evidence, traceability and decisions.
TOTAL	100%	Official weights total 100%.

Sprint 4: plan, order and handovers

SPRINT GOAL

30 September-11 October 2026

Finish claimed Advanced scope, harden and release

What this sprint must achieve

- Finish only the Advanced scope the team can demonstrate reliably: proposals, locks, collaboration/conflicts, scenarios, swaps, audit and restore.
- Harden the database, API and app with full regression, coverage, accessibility, performance and failure checks.
- Merge the tested release from develop to protected main and tag it.

Equal technical allocation

MEMBER	TECHNICAL OWNERSHIP	ISSUES	POINTS
Member 1	Authentication, role security, notifications/search and shared release checks	S4-M1-1 to S4-M1-3	12
Member 2	Organiser interface, allocation board, approvals, reports and scenarios	S4-M2-1 to S4-M2-3	12
Member 3	Tutor dashboard, availability, timesheets, excuses, imports and swaps	S4-M3-1 to S4-M3-3	12
Member 4	Student overflow plus shared responsive and accessible interface	S4-M4-1 to S4-M4-3	12
Member 5	Handwritten backend HTTP API, validation, business rules, security and API tests	S4-M5-1 to S4-M5-3	12
Member 6	Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery	-1 to S4-M6-3	12
TOTAL	Meetings and documentation are owned by the Project Coordinator.	18 issues	72

BEGINNER RULE

A contract is a written example of the data, API request, response and error shape. Once the example is agreed, frontend work may begin with a mock. A member waits only for the exact missing contract - not for the whole backend or whole sprint.

Sprint 4: who goes first and how work joins

WHO STARTS FIRST

Member 6 starts by freezing Advanced engine, data, lock/audit and recovery contracts. Member 5 hardens the API contract by contract. Members 1-4 begin final UI and edge-case tests with approved mocks. New feature work stops at the agreed freeze so everyone can help release safely.

Detailed order, parallel work and handover

STEP / WHEN	OWNER	DO THIS	SAFE PARALLEL WORK	HAND OVER TO	READY WHEN
1 Day 1	Member 6	Freeze Advanced engine/data, lock, audit, restore and rollback contracts; publish release checklist.	M5 prepares hardened routes. M1-4 list final UI/edge cases.	Member 5 + all feature owners	Contracts and the feature-freeze boundary are written and agreed.
2 Per ready contract	Member 5	Build/harden Advanced endpoints with conflict, permission, validation and recovery tests.	M6 verifies transactions, indexes, backup/restore and deployment behavior.	Relevant UI owner	API tests cover normal, forbidden, conflict and failure paths.
3 Same time	Members 1-4	Finish Advanced screens and final role edge cases with mocks; test keyboard, focus and common phone widths.	Each member owns their role area; no one waits for all endpoints.	Member 5	UI tests pass and every unresolved API need is an assigned blocker.
4 As hardened routes land	Members 1-4	Replace mocks, run full real journeys and verify conflict/retry messages preserve user work.	M5 fixes API issues; M6 checks data consistency and logs.	Issue reviewers	Advanced journey passes and Basic/Intermediate regression remains green.
5 Feature freeze	All members	Stop optional features. Fix only release defects, add a regression test and update the issue evidence.	Different defect areas can run in parallel; shared-file changes need one owner.	Integration Lead	No open severity-one/two defect; lower risks are documented.
6 Release rehearsal	Members 1, 5 + 6	Run security, API, migration, performance, external-failure, backup/restore, CI and Codecov gates.	Members 2-4 run role, accessibility and responsive regression.	Integration Lead	Fresh-environment setup and complete regression both pass.
7 Final release	Integration Lead	Merge reviewed PRs to develop one at a time, rerun all checks, then merge protected develop to main and tag.	Coordinator completes evidence and demonstration script; members remain available for fixes.	Project Coordinator + team	Tagged release, rollback notes and final evidence are complete.

How to hand over: post the contract or PR link in the dependent GitHub issue; include setup steps, sample data/request/response, tests run and known limits; tag the named receiver. The receiver replies that they can use it or opens a blocking issue naming the missing item. A handover is not complete just because code was pushed.

Sprint 4 - Member 1

YOUR SPRINT OWNERSHIP

Authentication, role security, notifications/search and shared release checks

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S4-M1-1 | Production account and session hardening

GitHub issue ID	S4-M1-1	Story points	4
Branch	s4/m1/auth-hardening	PR reviewer	Member 5
Labels	sprint-4, member-1, area-security, points-4	PR base	develop
Depends on	Production-like environment and full role matrix	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a Tutor Mx user, I want robust account/session behavior so that final release access is secure and understandable under edge cases.

Acceptance tests

- [] Given expired/revoked/missing session, when a protected action runs, then re-authentication is required without exposing data.
- [] Given reset/delete/sign-out boundaries, when completed, then stale tabs/tokens cannot continue protected work.
- [] Given wrong role/direct API request, when attempted, then 403/401 behavior remains consistent.
- [] Given an auth service error, when it occurs, then the user receives a safe recovery message and no false success.

Tasks

- [] Audit all account/session flows in production-like build.
- [] Fix stale-session, redirect-loop and wrong-role edge cases.
- [] Review browser storage and remove sensitive custom persistence.
- [] Expand security regression for expired/revoked tokens and stale tabs.
- [] Run coverage and production smoke for all three roles.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Member 5 fixes backend auth issues; Integration Lead requires green session regression.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S4-M1-2 | Notification/search final fixes and release smoke

GitHub issue ID	S4-M1-2	Story points	4
Branch	s4/m1/shared-release-smoke	PR reviewer	Member 5
Labels	sprint-4, member-1, area-release, points-4	PR base	develop
Depends on	Final feature scope and Coordinator's severity-one/two list	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want search, notifications and the public role entry paths smoke-tested so that final release does not hide critical breakage.

Acceptance tests

- [] Given clean Organiser/Tutor/Student accounts, when public app/API flows run, then entry, search and notifications work on the exact release candidate.
- [] Given missing/deleted notification target, when opened, then a safe state is shown.
- [] Given role-restricted search, when queried, then no forbidden result leaks.
- [] Given a severity-one/two defect in this area, when final release is proposed, then it is fixed/re-tested or release is blocked.

Tasks

- [] Run clean-browser smoke on account, search and notification paths.
- [] Fix assigned severity-one/two shared defects.
- [] Add regression tests for every fixed defect.
- [] Verify deployed URLs and environment configuration.
- [] Attach exact commit/tag candidate evidence to PR.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Integration Lead re-runs smoke after integration; Coordinator records release evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S4-M1-3 | Final auth/security regression and failure rehearsal

GitHub issue ID	S4-M1-3	Story points	4
Branch	s4/m1/security-regression	PR reviewer	Member 5
Labels	sprint-4, member-1, type-test, points-4	PR base	develop
Depends on	S4-M1-1, S4-M1-2 and final develop build	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the Integration Lead, I want a repeatable final auth/security failure rehearsal so that the release tag is not approved on verbal confidence alone.

Acceptance tests

- [] Given all required account actions and role routes, when final automated/manual regression runs, then every result is recorded against the release candidate.
- [] Given 401/403/cross-user cases, when tested, then no private response body is returned.
- [] Given temporary auth/network failure, when rehearsed, then recovery behavior is controlled.
- [] Given the final release tag, when checked, then the same tests/Codecov statuses are green.

Tasks

- [] Run and record the full auth/role/ownership test matrix.
- [] Rehearse one controlled auth/network failure.
- [] Open/fix/re-test defects assigned to Member 1.
- [] Verify Codecov patch/project status for Member 1 changes.
- [] Provide raw evidence to Coordinator; do not rewrite project documentation.
- [] Open a PR to develop and request Member 5 review.

Handoff / how this joins the sprint

Integration Lead uses this before develop-to-main release PR.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 4 - Member 2

YOUR SPRINT OWNERSHIP

Organiser interface, allocation board, approvals, reports and scenarios

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S4-M2-1 | Draft allocation scenarios, compare and publish

GitHub issue ID	S4-M2-1	Story points	4
Branch	s4/m2/scenario-publish	PR reviewer	Member 4
Labels	sprint-4, member-2, area-advanced, points-4	PR base	develop
Depends on	Member 5 scenario API and Member 6 engine/schema	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As an organiser, I want draft allocations side by side and an explicit publish action so that I can compare choices before changing live work.

Acceptance tests

- [] Given two drafts, when compared, then changed assignments, shortages, hours and objective summaries are visible.
- [] Given a draft, when edited, then live allocations do not change.
- [] Given locked assignments, when proposals run, then locks remain unchanged.
- [] Given authorised publish confirmation, when completed, then exactly one chosen scenario becomes live and users receive affected changes.

Tasks

- [] Build draft list/create/rename/delete and comparison UI.
- [] Clearly label draft versus live data.
- [] Show differences and lock status.
- [] Build explicit publish confirmation and result state.
- [] Add draft isolation, lock and publish tests.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Member 5/6 guarantee transactional publish; Member 1 notifications carry changes.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S4-M2-2 | Concurrent organiser presence and conflict resolution

GitHub issue ID	S4-M2-2	Story points	4
Branch	s4/m2/concurrent-board	PR reviewer	Member 4
Labels	sprint-4, member-2, area-advanced, points-4	PR base	develop
Depends on	Member 5 concurrency API and Member 6 version/audit data	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want to see other organisers and resolve conflicting edits so that one person's work is not silently overwritten.

Acceptance tests

- [] Given two organiser sessions, when both open the same board, then presence or last-update information is visible.
- [] Given both edit the same allocation/version, when the second save occurs, then a conflict is detected rather than silently overwriting.
- [] Given conflict, when resolved, then the organiser can refresh/compare/retry and the final chosen state is explicit.
- [] Given organiser disconnect, when presence expires, then stale presence does not remain indefinitely.

Tasks

- [] Build presence/last-update display.
- [] Send expected version with edits and handle conflict response.
- [] Build refresh/compare/retry conflict UI.
- [] Add two-session conflict and stale-presence tests.
- [] Check conflict flow works without relying only on colour.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Member 5/6 own version enforcement; Coordinator receives two-browser evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S4-M2-3 | Organiser performance, accessibility and final regression

GitHub issue ID	S4-M2-3	Story points	4
Branch	s4/m2/organiser-final-quality	PR reviewer	Member 4
Labels	sprint-4, member-2, type-test, points-4	PR base	develop
Depends on	S4-M2-1, S4-M2-2 and final organiser features	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As an organiser, I want the complete board to be fast, keyboard-usable and regression-tested so that the claimed tier is credible in production.

Acceptance tests

- [] Given realistic large data, when board, ranking, reports and scenario comparison load, then they meet the recorded target or a limitation is logged.
- [] Given keyboard-only use, when core allocation/bulk/compare/conflict actions run, then they complete successfully.
- [] Given Basic, Intermediate and claimed Advanced organiser paths, when regression runs, then all pass on release candidate.
- [] Given severity-one/two organiser defect, when release is proposed, then it is fixed/re-tested or blocks release.

Tasks

- [] Measure and improve assigned organiser performance bottleneck.
- [] Run keyboard/focus/label/status checks.
- [] Run full organiser regression including conflicts/locks/publish.
- [] Fix assigned final organiser defects and add regression tests.
- [] Attach coverage/performance evidence to PR.
- [] Open a PR to develop and request Member 4 review.

Handoff / how this joins the sprint

Integration Lead requires green organiser suite; Coordinator records claims/limits.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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Sprint 4 - Member 3

YOUR SPRINT OWNERSHIP

Tutor dashboard, availability, timesheets, excuses, imports and swaps

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S4-M3-1 | Constraint-checked tutor session swaps

GitHub issue ID	S4-M3-1	Story points	4
Branch	s4/m3/tutor-swaps	PR reviewer	Member 6
Labels	sprint-4, member-3, area-advanced, points-4	PR base	develop
Depends on	Member 5 swap API and Member 6 allocation engine/data	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As a tutor, I want to propose a session swap that is checked and approved so that changes do not break marks, timetable or weekly-hour rules.

Acceptance tests

- [] Given an eligible replacement tutor, when a swap is proposed, then all three constraints are checked for affected assignments.
- [] Given failed constraint, when proposed, then the swap is not applied and the reason is visible.
- [] Given a valid pending swap, when organiser approves/rejects, then both tutors see the outcome.
- [] Given no approval, when a proposal exists, then live assignments remain unchanged.

Tasks

- [] Build tutor swap proposal/select/reason UI.
- [] Show constraint results and pending status.
- [] Build or integrate organiser approval view.
- [] Add valid, invalid, rejected and no-early-change tests.
- [] Connect notifications for affected users.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 5/6 enforce rules/transaction; Member 2 integrates organiser approval if needed.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S4-M3-2 | Harden tutor, timetable and timesheet edge cases

GitHub issue ID	S4-M3-2	Story points	4
Branch	s4/m3/tutor-hardening	PR reviewer	Member 6
Labels	sprint-4, member-3, area-tutor, points-4	PR base	develop
Depends on	Full Tutor flow and final severity list	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As a tutor, I want boundary cases handled safely so that imports, hours, excuses and timesheets remain trustworthy at release.

Acceptance tests

- [] Given boundary dates/hours, duplicates and invalid imports, when submitted, then correct validation/history behavior occurs.
- [] Given returned/disputed/approved timesheet, when actions occur, then only allowed transitions are available.
- [] Given past/changed assignment excuse, when attempted, then the rule is explained and data stays consistent.
- [] Given final Tutor path, when run on common widths and keyboard only, then it completes.

Tasks

- [] Run boundary checklist across dashboard, import, exceptions, work logs, timesheets and excuses.
- [] Fix assigned severity-one/two tutor defects.
- [] Preserve status/history instead of silent replacement.
- [] Add regression tests for every final fix.
- [] Check keyboard, focus and phone layout.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Member 5/6 receive API/data defects; Coordinator receives final tutor evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S4-M3-3 | Complete tutor regression and payroll/export verification

GitHub issue ID	S4-M3-3	Story points	4
Branch	s4/m3/tutor-final-tests	PR reviewer	Member 6
Labels	sprint-4, member-3, type-test, points-4	PR base	develop
Depends on	S4-M3-1, S4-M3-2 and release candidate	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want the complete tutor workflow and payroll output verified so that final claims match deployed behavior.

Acceptance tests

- [] Given Basic, Intermediate and claimed Advanced tutor journeys, when final regression runs, then all required paths pass.
- [] Given known approved work, when payroll export runs, then fields/totals match fixtures and exclude unapproved data.
- [] Given cross-user/wrong-role requests, when run, then access is denied.
- [] Given the release tag, when CI/Codecov completes, then tutor tests and coverage status are green.

Tasks

- [] Run full tutor automated and clean-browser regression.
- [] Verify export schema/totals with known fixtures.
- [] Test swap approval plus all invalid constraint cases.
- [] Fix/re-test defects assigned to Member 3.
- [] Attach exact release-candidate evidence to PR.
- [] Open a PR to develop and request Member 6 review.

Handoff / how this joins the sprint

Integration Lead uses tutor suite before main; Coordinator writes final evidence from supplied results.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

Sprint 4 - Member 4

YOUR SPRINT OWNERSHIP

Student overflow plus shared responsive and accessible interface

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S4-M4-1 | Overflow race handling and notification polish

GitHub issue ID	S4-M4-1	Story points	4
Branch	s4/m4/overflow-hardening	PR reviewer	Member 2
Labels	sprint-4, member-4, area-student, points-4	PR base	develop
Depends on	Member 5 atomic claim API and Member 6 constraints	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As students and organisers, we want competing claims and status messages handled correctly so that only one valid outcome is accepted and everyone sees the truth.

Acceptance tests

- [] Given two students claiming the same item nearly together, when requests finish, then only the permitted winner/pending outcome exists.
- [] Given closed/filled work, when a stale page submits, then conflict is explained and list refreshes.
- [] Given approval/rejection/closure, when status changes, then only affected users receive correct notifications.
- [] Given missing target/late notification, when opened, then a safe current state is shown.

Tasks

- [] Run two-session claim race tests with Member 5/6 fixtures.
- [] Fix stale-page/conflict UI and refresh behavior.
- [] Remove duplicate/irrelevant notifications.
- [] Add regression tests for race, close, outcome and missing target.
- [] Check plain language and accessible status announcement.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Member 5/6 fix atomicity/constraint defects; Coordinator receives race-test evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S4-M4-2 | Final shared responsive and accessible visual system

GitHub issue ID	S4-M4-2	Story points	4
Branch	s4/m4/shared-polish	PR reviewer	Member 2
Labels	sprint-4, member-4, area-shared-ui, points-4	PR base	develop
Depends on	All final role screens integrated	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As any user, I want consistent, responsive and accessible screens so that Tutor Mx works without confusing visual or interaction differences.

Acceptance tests

- [] Given common phone/tablet/desktop widths, when every primary role page renders, then no critical control or information is clipped.
- [] Given keyboard-only use, when core journeys run, then focus is visible and logical.
- [] Given forms/statuses, when read by assistive technology, then labels, errors and changes are announced appropriately.
- [] Given final UI, when reviewed, then placeholder text/components and inconsistent duplicate patterns are removed.

Tasks

- [] Audit and fix shared navigation, forms, tables/cards and state components.
- [] Run contrast, label, focus and status-announcement checks.
- [] Run common-width screenshots/checks for all role pages.
- [] Fix assigned severity-one/two shared UI defects.
- [] Add regression tests for fixed shared failures.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Members 1-3 adopt shared fixes; Coordinator records accessibility/responsive evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S4-M4-3 | Student/shared UI final regression

GitHub issue ID	S4-M4-3	Story points	4
Branch	s4/m4/student-final-tests	PR reviewer	Member 2
Labels	sprint-4, member-4, type-test, points-4	PR base	develop
Depends on	S4-M4-1, S4-M4-2 and release candidate	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want student, notification, search and shared UI regression green so that the final app remains usable after all integrations.

Acceptance tests

- [] Given Basic/Intermediate student paths, when final regression runs, then list, claim, status and notification cases pass.
- [] Given slow/error responses, when tested, then loading/fallback/retry states remain understandable.
- [] Given keyboard/common widths, when shared regression runs, then core paths pass.
- [] Given the release tag, when CI/Codecov runs, then student/shared tests and required statuses are green.

Tasks

- [] Run full student/shared automated regression.
- [] Run clean-browser slow/error/fallback checks.
- [] Fix/re-test defects assigned to Member 4.
- [] Add tests for every final shared UI fix.
- [] Attach coverage and accessibility evidence to PR.
- [] Open a PR to develop and request Member 2 review.

Handoff / how this joins the sprint

Integration Lead uses this before main; Coordinator writes final evidence from supplied results.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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Sprint 4 - Member 5

YOUR SPRINT OWNERSHIP

Separate handwritten backend HTTP API, rules, security and API tests

3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S4-M5-1 | Advanced handwritten API workflows

GitHub issue ID	S4-M5-1	Story points	4
Branch	s4/m5/advanced-api	PR reviewer	Member 1
Labels	sprint-4, member-5, area-api, points-4	PR base	develop
Depends on	Member 6 advanced engine/schema and Basic/Intermediate API	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As the advanced frontend, I want protected scenario, conflict, swap and audit endpoints so that every claimed feature still uses the handwritten API.

Acceptance tests

- [] Given draft/compare/publish requests, when called, then drafts stay isolated and publish is authorised/transactional.
- [] Given stale version/conflicting edit, when saved, then 409-style conflict is returned without overwrite.
- [] Given swap proposal/approval, when called, then constraints and roles are enforced before live change.
- [] Given audit/restore request, when called, then history/permissions are enforced and the result is traceable.

Tasks

- [] Implement scenario/diff/publish endpoints.
- [] Implement version/conflict/presence support endpoints.
- [] Implement swap and audit/restore endpoints.
- [] Add permission, transaction, conflict and failure tests.
- [] Keep error contracts consistent and documented via raw examples for Coordinator.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Members 2/3 connect UI; Member 6 supplies engine/history transactions.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S4-M5-2 | API security, design and deployment hardening

GitHub issue ID	S4-M5-2	Story points	4
Branch	s4/m5/api-hardening	PR reviewer	Member 1
Labels	sprint-4, member-5, area-security, points-4	PR base	develop
Depends on	All final endpoints and production-like environment	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the school, we want a secure, available and maintainable API so that final deployment does not expose data or depend on development shortcuts.

Acceptance tests

- [] Given malformed/oversized/duplicate/unauthorised requests, when sent, then validation, idempotency/limits and permissions protect state.
- [] Given logs/errors, when inspected, then tokens, database credentials and private payloads are not exposed.
- [] Given deployment/rollback, when rehearsed, then the API remains externally available or can return to the prior known release.
- [] Given API contract, when checked, then versioning/status/error/pagination choices are consistent.

Tasks

- [] Audit all endpoints for auth, ownership, validation and secrets.
- [] Add pagination/limits/idempotency where required.
- [] Remove development bypasses and unsafe logs.
- [] Rehearse deploy/health/rollback with Member 6.
- [] Add security/deployment regression tests.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Integration Lead requires green security/deployment checks; Coordinator records architecture/deployment evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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S4-M5-3 | API load, availability and final regression

GitHub issue ID	S4-M5-3	Story points	4
Branch	s4/m5/api-final-tests	PR reviewer	Member 1
Labels	sprint-4, member-5, type-test, points-4	PR base	develop
Depends on	S4-M5-1, S4-M5-2 and realistic Member 6 data	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the team, we want measured API performance and complete regression so that final API rubric claims are supported by evidence.

Acceptance tests

- [] Given realistic load/data, when critical endpoints are measured, then results meet recorded targets or limitations are explicit.
- [] Given all Basic, Intermediate and claimed Advanced routes, when full regression runs, then success, boundary, permission and failure cases pass.
- [] Given dependency/database slow failure, when tested, then health/error behavior is controlled.
- [] Given release tag, when CI/Codecov runs, then backend suite and required statuses are green.

Tasks

- [] Run load/timing checks on allocation, search, report, scenario and publish routes.
- [] Fix assigned high-impact performance defects.
- [] Run full API/security/concurrency regression.
- [] Verify deployed health and external availability.
- [] Attach measured results and coverage to PR.
- [] Open a PR to develop and request Member 1 review.

Handoff / how this joins the sprint

Integration Lead uses this before main; Coordinator records API performance/testing evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
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END OF COPY

Sprint 4 - Member 6

YOUR SPRINT OWNERSHIP

Neon PostgreSQL, Prisma migrations, external integration, allocation engine, CI/Codecov and delivery
3 issues x 4 points = 12 technical points. Work from top to bottom unless the dependency notes say work can begin with the agreed mock.

S4-M6-1 | Explainable school-wide allocation engine with locks

GitHub issue ID	S4-M6-1	Story points	4
Branch	s4/m6/advanced-allocation-engine	PR reviewer	Member 3
Labels	sprint-4, member-6, area-advanced, points-4	PR base	develop
Depends on	Final constraints/objectives confirmed by Coordinator/client	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY

User story

As an organiser, I want a school-wide proposal that respects constraints/locks and explains choices so that I can trust and evaluate the recommendation.

Acceptance tests

- [] Given courses, tutors and constraints, when engine runs, then it produces a proposal or explicit infeasibility without breaking hard rules.
- [] Given locked assignments, when engine runs, then locks remain unchanged.
- [] Given each proposed assignment, when explanation is requested, then chosen factors are stated.
- [] Given an eligible tutor not chosen, when explanation is requested, then a reason/non-selection factor is stated without pretending certainty.

Tasks

- [] Implement deterministic proposal engine with separate hard constraints and objectives.
- [] Preserve organiser locks and report unmet work.
- [] Generate structured chosen/not-chosen explanations.
- [] Add tests for feasible, impossible, tie, lock and competing-objective cases.
- [] Measure realistic engine time and tune safely.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Member 5 exposes engine; Member 2 renders proposals/explanations.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S4-M6-2 | Audit history, restore and production database recovery

GitHub issue ID	S4-M6-2	Story points	4
Branch	s4/m6/audit-restore	PR reviewer	Member 3
Labels	sprint-4, member-6, area-database, points-4	PR base	develop
Depends on	All final mutation paths and Member 5 audit endpoints	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the school, we want every important change recorded and restorable so that the system can explain who changed what and recover from mistakes.

Acceptance tests

- [] Given allocation/timesheet/claim/scenario changes, when committed, then actor, time, action and before/after reference are recorded.
- [] Given authorised restore, when an earlier allocation version is selected, then restoration creates a new auditable change rather than deleting history.
- [] Given database backup/recovery procedure, when rehearsed in a safe environment, then required data returns consistently.
- [] Given unauthorised restore/history access, when attempted, then it is rejected.

Tasks

- [] Complete audit event/version schema and transaction hooks.
- [] Implement safe allocation restore service/data operations.
- [] Add indexes/retention choices needed for audit queries.
- [] Rehearse backup/restore or documented recovery mechanism in non-production environment.
- [] Add history, restore, permission and recovery tests.
- [] Open a PR to develop and request Member 3 review.

Handoff / how this joins the sprint

Member 5 exposes authorised endpoints; Coordinator receives recovery/audit evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

END OF COPY

S4-M6-3 | Release pipeline, Codecov gate and integration resilience

GitHub issue ID	S4-M6-3	Story points	4
Branch	s4/m6/final-release-gate	PR reviewer	Member 3
Labels	sprint-4, member-6, type-test, points-4	PR base	develop
Depends on	All final code on develop and Integration Lead access	Merge	Integration Lead only

COPY FROM HERE INTO THE GITHUB ISSUE BODY**## User story**

As the Integration Lead, I want a repeatable final release gate so that only the reviewed, tested and deployable commit reaches main.

Acceptance tests

- [] Given release PR from develop to main, when opened, then all required lint/test/coverage/security/smoke checks run and pass before merge.
- [] Given failed migration, external outage or health check, when rehearsed, then release is blocked or controlled fallback/rollback succeeds.
- [] Given Codecov, when final PR runs, then frontend/backend flags and project/patch statuses are visible.
- [] Given the main release, when tagged/deployed, then the tag matches the tested commit and public app/API/docs links are reachable.

Tasks

- [] Run clean migration, full tests, coverage and production-like smoke.
- [] Verify branch protection and required checks on develop/main.
- [] Rehearse external failure and deploy/rollback behavior.
- [] Fix flaky CI, secret leakage or path errors.
- [] Provide exact run/tag/link evidence to Coordinator.
- [] Open a PR to develop and request Member 3 review; Integration Lead alone opens develop-to-main PR.

Handoff / how this joins the sprint

Integration Lead executes the final develop-to-main merge; Coordinator writes submission evidence.

Definition of done

- [] All acceptance tests above pass and no new severity-one/two defect is left open.
- [] Local tests and coverage were run; CI and Codecov checks are green.
- [] PR targets develop, links this issue using Closes #NUMBER, and has the assigned review.
- [] No secrets or personal data are committed; Tutor Mx data goes through the handwritten Express API; the browser never connects directly to Neon.
- [] If AI generated code, relevant commits include Assisted-by: <tool>[<model>]; the Coordinator is told if README AI declarations must change.

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Sprint 4: integration review and official rubric

OFFICIAL DESCRIPTION NOTE

The latest project brief publishes the Sprint 3 and final criterion names/weights but leaves Advanced/Intermediate/Basic description cells blank. The evidence wording below is team readiness guidance, not a replacement for the lecturer's rubric.

Integration Lead release gate

- [] Every issue is linked to a reviewed PR targeting develop; no Member pushes directly to main.
- [] The assigned reviewer checks the issue's acceptance tests, changed code, local/CI tests, coverage and AI attribution.
- [] Integration Lead merges approved PRs into develop one at a time, resolves conflicts with the author, then runs the full suite.
- [] Only the Integration Lead opens develop -> main. Main remains protected and merges only after required checks pass.
- [] The Project Coordinator runs meetings, maintains the GitHub board/decisions and writes project documentation from Member evidence.

Sprint review demonstration checklist

- [] Start from public production links and clean accounts.
- [] Demonstrate every claimed tier feature; do not claim unimplemented Advanced features.
- [] Show explanations/locks/scenarios/conflicts/swaps/audit/restore if claimed.
- [] Show accessibility, responsiveness, performance, recovery and integration failure evidence.
- [] Show final main tag matches the tested commit and all checks are green.

Official rubric weights

Database: Data	3%	Correct, suitable production-like data handling.
Database: Deployment	2%	Repeatable deployed data setup/recovery.
Database: Structure	5%	Sound keys, constraints, migrations and indexes.
API: Availability	3%	Externally reachable and reliable.
API: Architecture	5%	Separate handwritten API and sound boundaries.
API: Deployment	2%	Repeatable secure deployment/rollback.
API: Performance	5%	Measured/tuned critical endpoints.
API: Design	10%	Consistent, secure, usable HTTP contract.
App: Accessibility	5%	Keyboard, labels, focus, contrast and status behavior.
App: Aesthetics	3%	Coherent finished visual presentation.
App: User Experience	5%	Role tasks are clear and efficient.
App: Deployment	2%	Public working app release.
App: Performance	5%	Measured/tuned critical UI journeys.
App: Features	10%	Claimed tier works end to end.
App: Responsiveness	5%	Core tasks work at agreed widths.
App: Structure	5%	Maintainable separate frontend structure.

Misc: Git Methodology	5%	Traceable protected PR workflow.
Misc: Integration	7%	Frontend, Express API, Neon PostgreSQL and external service work together.
Misc: Testing	8%	Complete useful regression and failure evidence.
Misc: Tools	5%	Appropriate development/quality tools used well.
TOTAL	100%	Official weights total 100%.

Requirement-to-issue traceability

This matrix shows where each official requirement first becomes usable and where it is hardened. It helps the Project Coordinator create documentation and review evidence without assigning documentation writing to Members.

Separate frontend/backend + handwritten API	S1-M5-1, S1-M5-2	S2-M5-1..3; S4-M5-1..3
Auth0: sign-up/sign-in/reset/delete	S1-M1-1, S1-M1-2	S2-M1-1..3; S4-M1-1..3
Neon PostgreSQL + Prisma migrations/data	S1-M6-1	S2-M6-1; S3-M6-1; S4-M6-2
CI/CD, GitHub and Codecov	S1-M6-2	S2-M6-3; S3-M6-3; S4-M6-3
Responsive/accessibility	S1-M4-2	S2-M4-3; S3-M4-2..3; S4-M4-2..3
Relevant external API + fallback	S1-M6-3	S2-M6-2/S2-M4-3; S4-M6-3
Basic organiser allocation board + 3 checks	S1-M2-1/S1-M2-3/S1-M5-3	S2-M2-1..3/S2-M5-1
Basic tutor dashboard/log/timesheet/excuse	S1-M3-1..3	S2-M3-1..3/S2-M5-2
Basic student overflow volunteering	S1-M4-1/S1-M4-3	S2-M4-1..2/S2-M5-2
Intermediate ranking/bulk/shortage/override	S3-M2-1..2	S3-M5-1/S3-M6-2; S4 regression
Intermediate timetable/import/exceptions	S3-M3-1	S3-M5-2/S3-M6-1; S4-M3-2
Intermediate corrections/disputes/export	S3-M3-2	S3-M5-2/S3-M6-1; S4-M3-3
Intermediate claims/notifications/reports/search/onboarding	S3-M1-1..2/S3-M2-3/S3-M4-1	S3-M5-2/S3-M6-1..2; S4 quality
Advanced explainable school-wide allocation + locks	S4-M6-1	S4-M5-1/S4-M2-1
Advanced collaboration/conflicts/scenarios/publish	S4-M2-1..2	S4-M5-1/S4-M6-2
Advanced tutor swaps	S4-M3-1	S4-M5-1/S4-M6-1
Advanced audit and restore	S4-M6-2	S4-M5-1

PUBLIC DOCUMENTATION REMAINS REQUIRED

The Project Coordinator owns the documentation work, but Members must provide accurate raw evidence: issue/PR links, commands, screenshots, API examples, test/coverage results, limitations and AI-attribution changes. A Member must not mark an issue Done if the evidence handoff is missing.

Official dates and source references

Official project dates

Sprint 1 review	25 August 2026
Sprint 2 review	15 September 2026
Sprint 3 review	29 September 2026
Final system submission	11 October 2026
Group report	18 October 2026
Group presentation	19-23 October 2026
Individual report and peer review	27 October 2026

Project source files

- project_briefs.pdf - official COMS3011A dates, key requirements and rubric weights.
- tutor_management.pdf - official Tutor Management System Basic, Intermediate and Advanced feature brief.
- ai_policy.pdf - required document, README, commit and assessment AI attribution rules.
- The earlier Tutor Mx sprint handbook - structure and retained six-Member continuity only; this edition replaces its task allocation.

Current technical references checked for this edition

- Codecov: GitHub uploads and frontend/backend flags
- GitHub: protected branches and required status checks
- Express.js: routing and middleware
- Auth0 React SDK: login/logout and access tokens
- Auth0: password reset and user management
- Neon + Prisma: PostgreSQL connection, pooling and migrations

FINAL HUMAN CHECK

Before copying issues into GitHub, use this stack unless the tutor tells the group to change it: React + Vite frontend; Node.js + Express handwritten API; Neon PostgreSQL; Prisma; Auth0 authentication; GitHub Actions + Codecov. The group still must confirm the external API choice, exact mark threshold/override rules and Member identities.

AI Declaration: This revised edition was updated with ChatGPT [GPT-5.6 Sol].